ASESII 24A02 Final Project

Predict noise level generated by NACA airfoil blade sections

Group 02
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Contents

Authorship and contribution statement

Student	ID	Main contributions
Nguyen Thi Yen Nhi	24125071	Target and predictors, split and preprocessing plan, and data exploration
Nguyen Khang Thinh	24125104	Dataset selection, research questions, hypotheses, and descriptive statistics for the experimental design
Nguyen Van Tinh	24125106	OLS, Ridge, and Lasso regression, including model fitting, regularization, coefficient analysis, and model comparison
Nguyen Le Quoc Huy	24125100	ANOVA/factorial design analysis, post-hoc analysis, and conclusion and evaluation

Abstract

This study investigates the prediction of sound pressure levels generated by NACA 0012 airfoil blade sections using five aerodynamic and geometric predictors. The dataset contains 1,503 wind-tunnel observations with no missing values, duplicate rows, or physically invalid measurements. Exploratory analysis revealed weak individual predictor-response relationships and moderate correlation between attack angle and suction-side displacement thickness, while the frequency-response relationship showed evidence of nonlinearity.

We compared Ordinary Least Squares (OLS), Ridge, and Lasso regression using an 80/20 train-test split and five-fold cross-validation. Predictors were standardized using training-set statistics only to prevent data leakage. OLS achieved a cross-validation MSE of 23.03, while Ridge and Lasso produced similar predictive errors. However, Ridge substantially improved numerical conditioning, reducing the condition number from 12.30 for OLS to 5.34 at its optimal penalty. Lasso did not eliminate any of the five original predictors at either tuning rule.

A perturbation experiment introduced a near-duplicate predictor highly correlated with frequency (r=0.995) to evaluate model stability under stronger multicollinearity. Ridge maintained stable predictive performance by redistributing coefficients across correlated predictors, whereas Lasso demonstrated variable-selection behavior under the more conservative 1SE rule.

Overall, the results show that regularization provides greater numerical stability without sacrificing substantial predictive accuracy, with Ridge offering the most consistent response to multicollinearity in this dataset.

1 Prediction design and leakage control

1.1 Target and predictors

```
data <- read.table("data/airfoil_self_noise.dat", header = FALSE)
colnames(data) <- c("frequency", "attack_angle", "chord_length",
↪  "free_stream_velocity", "suction_side_displacement_thickness",
↪  "scaled_sound_pressure")
```

```r
# Always (re)written from the .dat source so the cached CSV can never drift
# from the column-naming convention fixed above.
write.csv(
  data,
  "data/airfoil_self_noise.csv",
  row.names = FALSE
)

str(data)
```

```
## 'data.frame':    1503 obs. of  6 variables:
##  $ frequency                         : int  800 1000 1250 1600 2000 2500 3150
↪  4000 5000 6300 ...
##  $ attack_angle                      : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ chord_length                      : num  0.305 0.305 0.305 0.305 0.305 ...
##  $ free_stream_velocity              : num  71.3 71.3 71.3 71.3 71.3 71.3
↪  71.3 71.3 71.3 71.3 ...
##  $ suction_side_displacement_thickness: num  0.00266 0.00266 0.00266 0.00266
↪  0.00266 ...
##  $ scaled_sound_pressure             : num  126 125 126 128 127 ...
```

```r
config <- list(
  file = "airfoil_self_noise.csv",
  response = "scaled_sound_pressure",
  excluded = "scaled_sound_pressure"
)
data_raw <- read.csv(find_exam_data(config$file), check.names = FALSE)
predictors <- setdiff(names(data_raw), config$excluded)
analysis_data <- data_raw[, c(config$response, predictors), drop = FALSE]

if (anyNA(analysis_data)) stop("Declare a training-only missing-value plan.")
if (!all(vapply(analysis_data, is.numeric, logical(1)))) {
  stop("The assigned response and predictors must be numeric.")
}
```

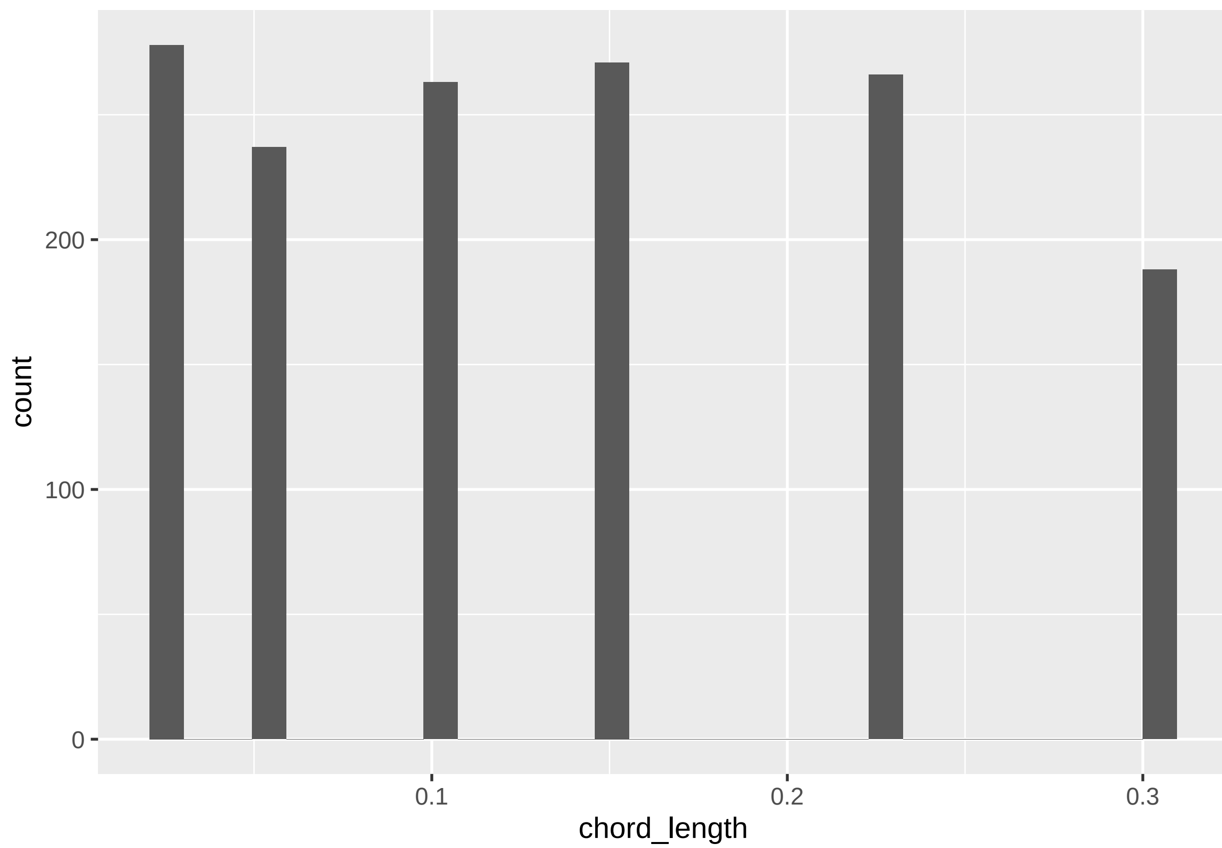


Table 1: Response and candidate predictors, with the integrity checks run before the split.

Variable	Role	Type	Distinct	Missing
frequency	Predictor	integer	21	0
attack_angle	Predictor	numeric	27	0
chord_length	Predictor	numeric	6	0
free_stream_velocity	Predictor	numeric	4	0
suction_side_displacement_thickness	Predictor	numeric	105	0
scaled_sound_pressure	Response	numeric	1456	0

Table 2: Missing-value and duplicate-row checks on the raw data.

Check	Count
Missing values (total cells)	0
Fully duplicated rows	0

The dataset contains 1503 observations of 6 numeric variables: five predictors recorded during NACA 0012 airfoil wind-tunnel tests and the response `scaled_sound_pressure`, the scaled sound pressure level in decibels. No missing values or duplicated rows are present (table above), so no imputation or de-duplication is required before the split.

The `Distinct` column in Table 1 shows that the five predictors are not equally "continuous" in practice. `frequency`, `chord_length`, and `free_stream_velocity` take only 21, 6, and 4 distinct values respectively - these are the fixed 1/3-octave frequency bands, airfoil chord lengths, and wind-tunnel speeds used to design the experiment, so each behaves more like an ordinal experimental factor than a freely varying measurement. `attack_angle` (27 distinct values) and `suction_side_displacement_thickness` (105 distinct values) are closer to genuinely continuous quantities. This low cardinality on three of the five predictors is expected to show up later as visibly clustered histograms and boxplots (1C) rather than smooth continuous distributions, and is a property of the experimental design rather than a data-quality issue.

`data/data_description.md` (the accompanying UCI variable sheet) lists `attack_angle` as "Binary", which does not match what is observed here - the variable takes 27 distinct values between 0 and 22.2 degrees. This looks like a labelling error in the source sheet, and `attack_angle` is treated as continuous throughout, consistent with the assignment brief's own description of the dataset ("angle of attack" as a continuous input).

1.2 Split and preprocessing plan

```r
split <- split_rows(nrow(analysis_data), seed = seeds$split)
train_id <- split$train
test_id <- split$test
train_df <- analysis_data[train_id, ]

x_raw <- as.matrix(analysis_data[, predictors, drop = FALSE])
y_raw <- analysis_data[[config$response]]

x_prep <- fit_scaler(x_raw[train_id, , drop = FALSE])
x_train <- apply_scaler(x_raw[train_id, , drop = FALSE], x_prep)
x_test <- apply_scaler(x_raw[test_id, , drop = FALSE], x_prep)
y_train <- y_raw[train_id]
y_test <- y_raw[test_id]

foldid <- make_foldid(nrow(x_train), k = 5L, seed = seeds$cv)
```

To ensure a fair comparison among OLS, ridge, lasso, and elastic net, all preprocessing and model development steps were performed using only the training data. The holdout test set was reserved exclusively for the final evaluation and was not used during model fitting, model selection, or hyperparameter tuning.

1.2.1 Data split

The observations were randomly partitioned into training and test sets using the custom `split_rows()` function with a training proportion of 80%. For Group 02, the split seed was **240202**, ensuring that the same partition can be reproduced in future analyses.

Observation	Value
Training	1202
Test observations	301

```
##
##  Chi-squared test for given probabilities
##
## data:  obs
## X-squared = 6, df = 3, p-value = 0.1116
```

The training data were used for exploratory analysis, preprocessing, model fitting, and cross-validation, while the holdout test data remained untouched until the final evaluation.

1.2.2 Predictor standardization

Since ridge regression, lasso, and elastic net are sensitive to the scale of predictor variables, all predictors were standardized before model fitting. The centering and scaling parameters were estimated **only from the training data**, and these same values were subsequently applied to the test predictors.

Using training-derived scaling parameters prevents information from the holdout observations from influencing the preprocessing stage and ensures that the test set remains an unbiased evaluation dataset.

1.2.3 Five-fold cross-validation

Model tuning was carried out using five-fold cross-validation on the training data only. Fold assignments were generated using the fixed cross-validation seed **240302** and were shared across all regularized models. Reusing identical folds ensures that differences in cross-validation performance are attributable to the models themselves rather than random variation in fold assignment.

1.2.4 Missing-value handling

The dataset was checked for missing values before model fitting.

Total_missing_values
0

No missing values were detected; therefore, no imputation was required. If missing values had been present, imputation statistics would have been estimated using only the training data and then applied unchanged to the test data.

1.2.5 Invalid-value check

Table 3: Physical-plausibility checks on the training predictors.

Variable	Rule	Violations
frequency	> 0 Hz	0
attack_angle	>= 0 deg	0
chord_length	> 0 m	0
free_stream_velocity	> 0 m/s	0
suction_side_displacement_thickness	> 0 m	0

Beyond the statistical IQR screen below, each predictor was also checked against the range it is physically allowed to take: a frequency, chord length, velocity, or boundary-layer thickness cannot be zero or negative, and an angle of attack cannot be negative. No violations are found on the training set, which is expected for an instrument-recorded wind-tunnel experiment rather than manually-entered data.

1.2.6 Outlier detection and handling

Table 4: Training-set values flagged by the 1.5×IQR rule, computed on the training split only.

Variable	Outliers	Percent
frequency	72	6.0
attack_angle	23	1.9
chord_length	0	0.0
free_stream_velocity	0	0.0
suction_side_displacement_thickness	111	9.2
scaled_sound_pressure	3	0.2

Outliers were screened with the standard 1.5 × IQR fence, applied to each variable using **training-set quantiles only** so that no information from the held-out test rows enters the check. `chord_length` and `free_stream_velocity` contribute zero flagged points, because - as noted in 1A - each takes only a handful of fixed values, so every observation sits well inside its own IQR fence by construction. `frequency`, `attack_angle`, and `suction_side_displacement_thickness` each contribute a nontrivial share (72, 23, and 111 training rows respectively), and 3 response values are flagged.

These flagged points are **retained rather than removed**. The airfoil data come from a controlled wind-tunnel experiment, not a survey or manual data-entry process, so there is no mechanism here for typographical or transcription errors; every extreme value is a physically valid test condition (a high frequency band, a large angle of attack, a thin boundary layer) rather than a data-quality defect. The IQR rule is doing exactly what it is designed to do on a right-skewed engineering variable: flagging the upper tail of a real physical distribution, not finding mistakes. Dropping these rows would bias the training sample away from precisely the loud, high-frequency regime the model is meant to predict, so all 1202 training rows are kept for model fitting in Problem 2.

1.2.7 Transformation check

Table 5: Training-set skewness of each predictor, and after a log transform for the two most skewed variables.

Variable	Skewness	Skewness_after_log
frequency	2.183	-0.010
attack_angle	0.695	NA
chord_length	0.452	NA
free_stream_velocity	0.259	0.013
suction_side_displacement_thickness	1.719	NA

`frequency` and `suction_side_displacement_thickness` are the two right-skewed predictors identified above (skewness 2.18 and 1.72). A log transform brings both close to symmetric on the training set (skewness -0.01 and 0.01), which is the usual motivation for log-transforming a positive, right-skewed engineering measurement. `attack_angle` is also mildly right-skewed (skewness 0.69, Table above) but far less than `suction_side_displacement_thickness`, so it is not treated as a transform candidate here.

However, symmetrizing the marginal distribution is not the same as improving the fit, and it is checked rather than assumed. Refitting a plain linear model on the training data with these two predictors log-transformed gives $R^2 = 0.4665$, against $R^2 = 0.5138$ for the untransformed predictors - the log transform does not improve, and mildly *hurts*, the linear fit. A separate check on `frequency` alone points to why: a quadratic term explains more of the response than a linear term ($R^2 = 0.1814$ vs. $R^2 = 0.1625$), so the frequency-response relationship is curved in a way a log transform does not match - consistent with the assignment brief's own framing of this dataset as having “few features but nonlinear relationships.”

Given this evidence, no log transform is applied. All predictors are carried forward on their original scale into Problem 2, with only the training-derived standardization described above; the nonlinearity documented here is noted as a limitation of the linear and regularized-linear models used there, rather than something a simple transform resolves.

1.2.8 Data cleaning summary

Table 6: Summary of data-cleaning checks and decisions carried into Problem 2.

Check	Result	Decision
Missing values	0 cells	No imputation needed

Check	Result	Decision
Duplicate rows	0 rows	No de-duplication needed
Invalid values	0 violations	No rows dropped
IQR outliers	209 variable-level flags (a row may be flagged on more than one variable)	Retained (valid experimental conditions, not errors)
Skewness	frequency and suction_side_displacement_thickness right-skewed	No log transform applied (does not improve the linear fit)
Categorical variables	none present (all 5 predictors numeric)	No encoding needed
Predictor scaling	required for ridge / lasso / elastic net	Applied in Problem 2, fit on training data only

1.2.9 Leakage control

Several safeguards were implemented to prevent information leakage throughout the analysis:

- The train–test split was performed before any preprocessing.
- Predictor standardization used statistics estimated from the training data only.
- The invalid-value, outlier, and transformation checks above used training-set values, quantiles, and model fits only.
- Five-fold cross-validation was conducted exclusively within the training set.
- Hyperparameter tuning for ridge, lasso, and elastic net relied solely on training cross-validation error.
- The holdout responses (`y_test`) were not accessed until the final evaluation stage.

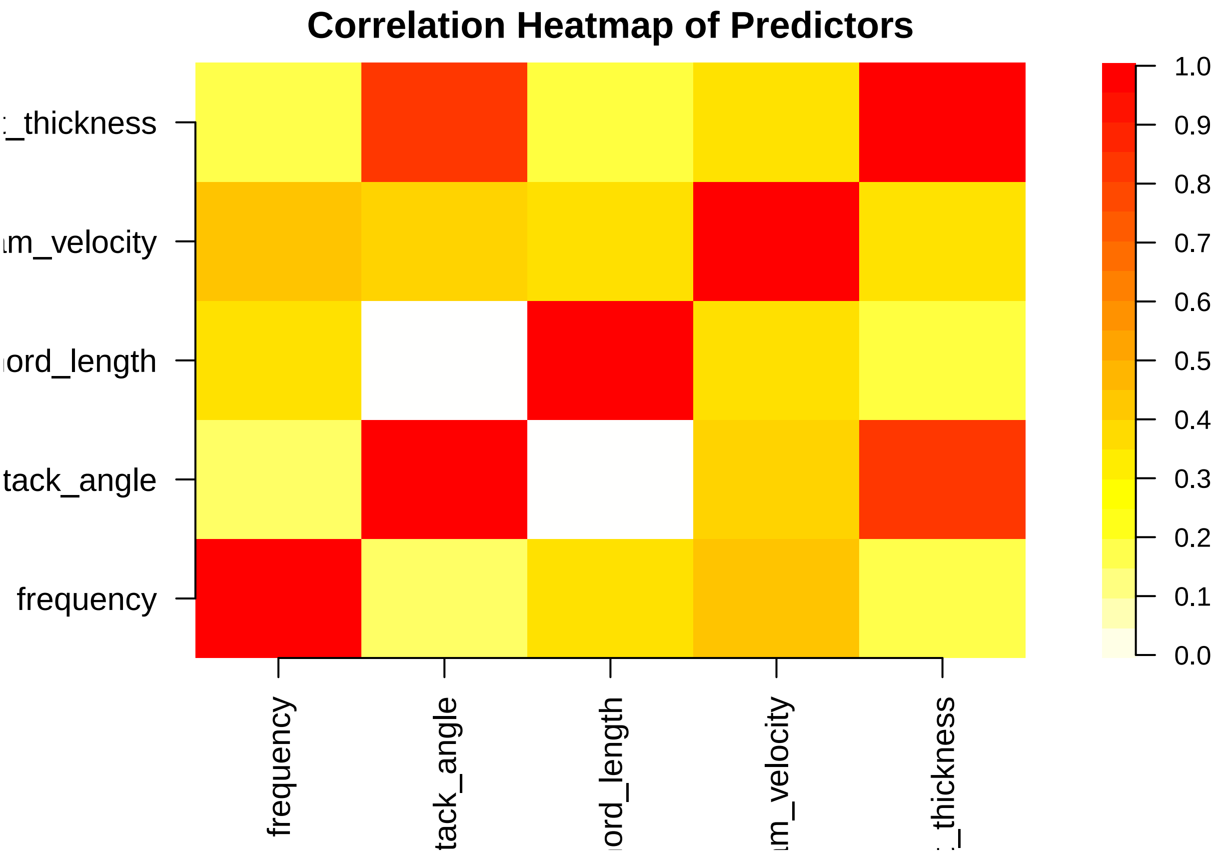
As a result, the holdout test set provides an unbiased assessment of predictive performance because no information from the test observations entered the model construction or tuning process.

1.3 Data exploration

1.3.1 Pairwise correlation among predictors

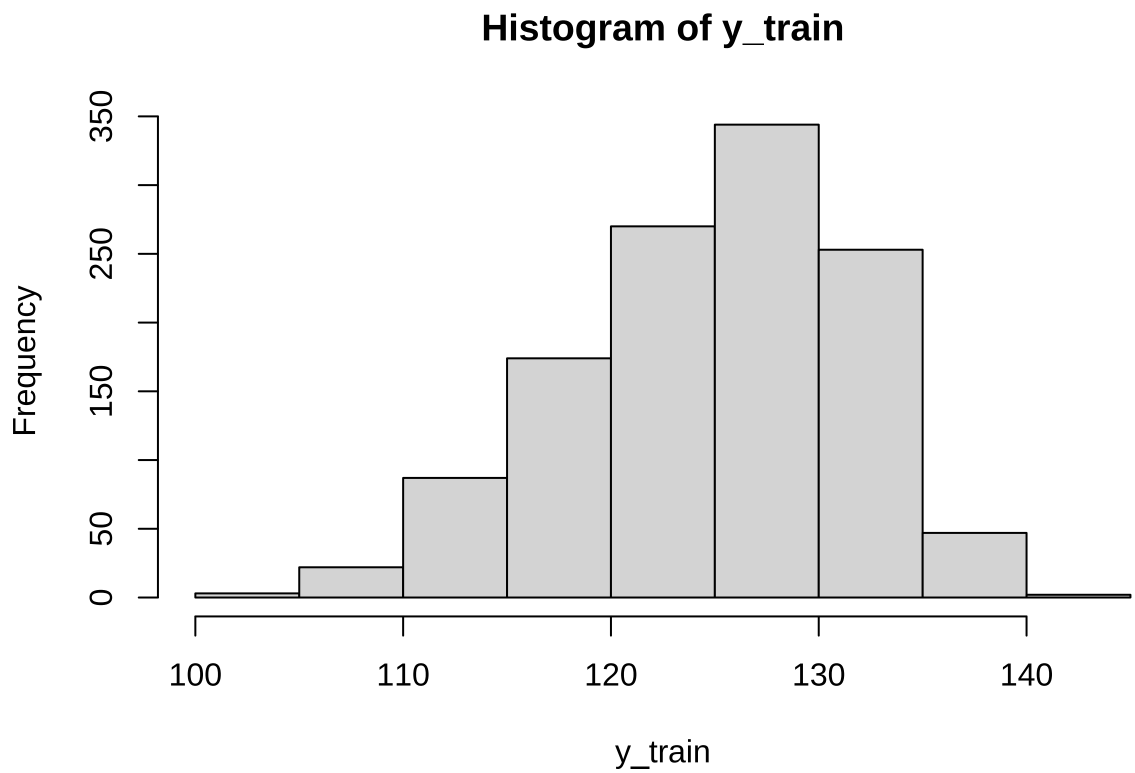
Table 7: Pairwise correlation matrix among predictors

	frequency	attack_angle	chord_length	free_stream_velocity	suction_side_displacement_thickness
frequency	1.00	-0.28	0.01	0.14	-0.24
attack_angle	-0.28	1.00	-0.50	0.07	0.76
chord_length	0.01	-0.50	1.00	0.01	-0.22
free_stream_velocity	0.14	0.07	0.01	1.00	0.00
suction_side_displacement_thickness	-0.24	0.76	-0.22	0.00	1.00



Predictor-predictor correlations are weak across most pairs. The one exception is suction_side_displacement_thickness and attack_angle, correlated at $r \approx 0.76$ - a moderate relationship, plausible given both describe the airfoil's boundary-layer geometry at the trailing edge. No other pair exceeds $|r| \approx 0.5$. This single moderate correlation is revisited with the VIF check below and is the main multicollinearity signal this dataset offers ridge and lasso a reason to address in Problem 2.

1.3.2 Response Distribution

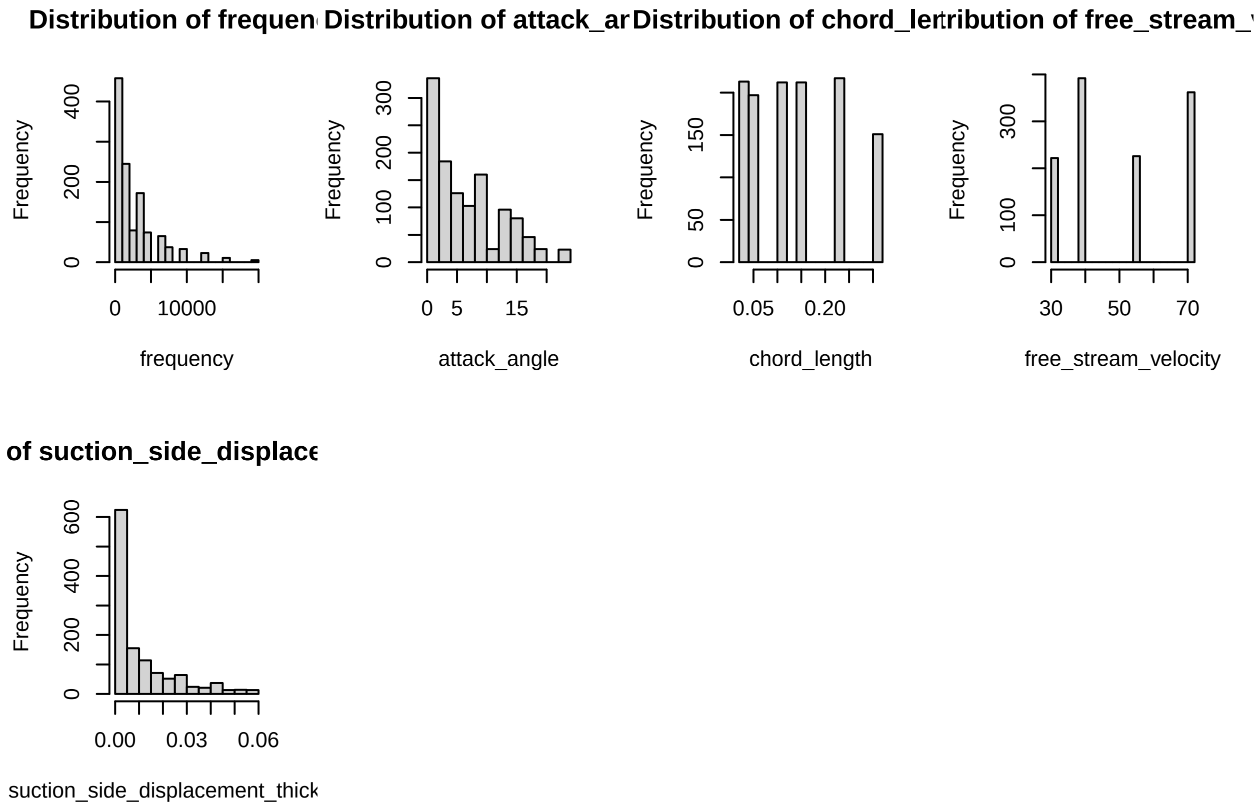


Row	Value
Min.	103.3800
1st Qu.	120.1900
Median	125.6385
Mean	124.8073
3rd Qu.	130.0312
Max.	140.9870

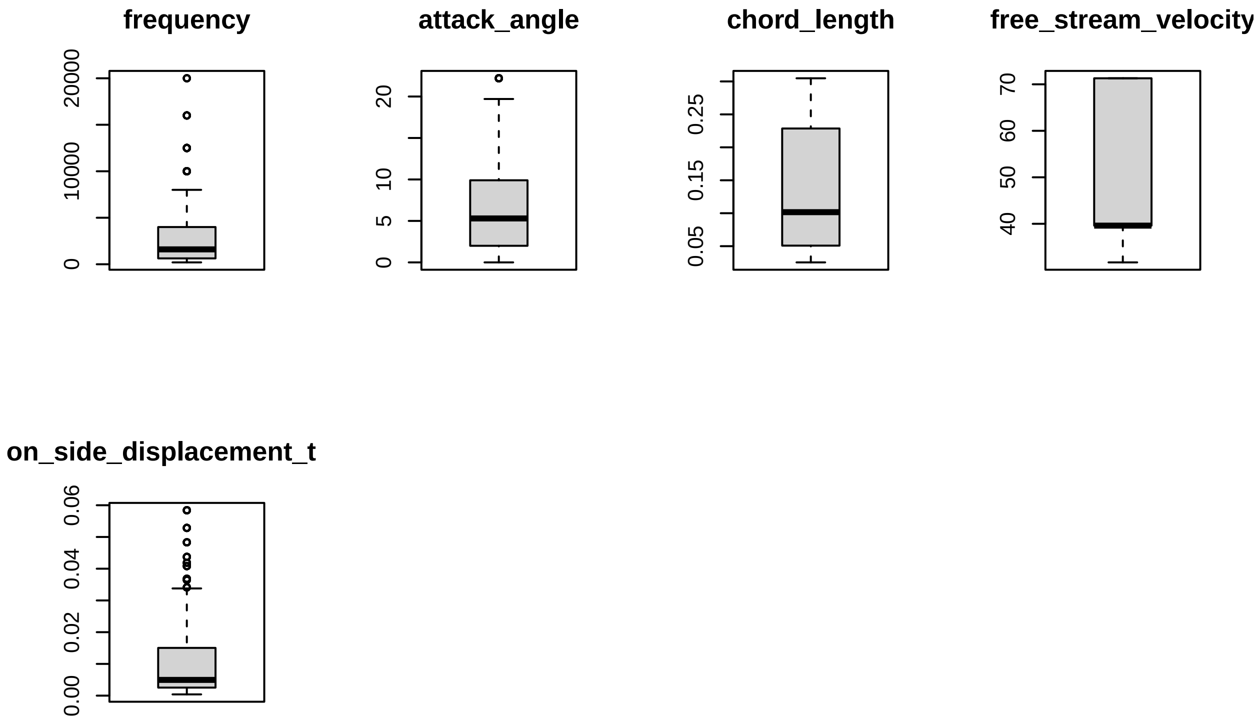
The response ranges from 103.4 to 141 dB with a median of 125.6 dB and mean 124.8 dB. The histogram and the small gap between mean and median show a mild left skew (skewness ≈ -0.42): most measurements cluster in the 120- 130 dB range, with a longer tail toward the quieter (lower-dB) end rather than the loud end. The skew is mild enough that no response transformation is applied - RMSE and MAE on the original dB scale remain directly interpretable as sound-pressure prediction errors, which matters for Problem 1's evaluation step.

1.3.3 Predictor Histogram & Boxplot

1. Histogram


Distribution of frequen
Frequency
0
200
400
0
10000
frequency
Distribution of attack_ar
Frequency
0
100
200
300
0
5
15
attack_angle
Distribution of chord_len
Frequency
0
50
150
0.05
0.20
chord_length
tribution of free_stream_v
Frequency
0
100
300
30
50
70
free_stream_velocity
of suction_side_displace
Frequency
0
200
400
600
0.00
0.03
0.06
suction_side_displacement_thick

2. Boxplot



The histograms and boxplots make the 1A distinct-value counts visible. `chord_length` and `free_stream_velocity` appear as 6 and 4 narrow, evenly spaced spikes rather than a smooth spread, and `frequency` shows 21 spikes concentrated at the low end with a long right tail - these are the fixed experimental levels described in 1A, not gaps in the data. `attack_angle` and `suction_side_displacement_thickness` are closer to continuous, right-skewed shapes, matching the skewness values quantified in 1B. The boxplots' upper whiskers and flagged points for these two variables are the same training-set values already counted in the 1B outlier check, retained for the same reason: they are valid high-frequency / high-angle test conditions, not data errors.

1.3.4 Correlation between predictors and response

Table 8: Correlation between predictors and scaled_sound_pressure

	scaled_sound_pressure
frequency	-0.40
attack_angle	-0.16
chord_length	-0.23
free_stream_velocity	0.11
suction_side_displacement_thickness	-0.30

Every predictor's marginal correlation with the response is weak in magnitude (all $|r| < 0.4$). frequency is the strongest single linear predictor ($r \approx -0.4$), followed by `suction_side_displacement_thickness`; `free_stream_velocity` is weakest and the only predictor positively correlated with the response. No individual predictor comes close to explaining the response on its own, which is consistent with the assignment brief's description of this dataset as having "few features but nonlinear relationships" - a linear combination of all five predictors, rather than any single one, is needed, and Problem 2's regularized models are expected to combine these weak individual signals rather than select one dominant predictor and discard the rest.

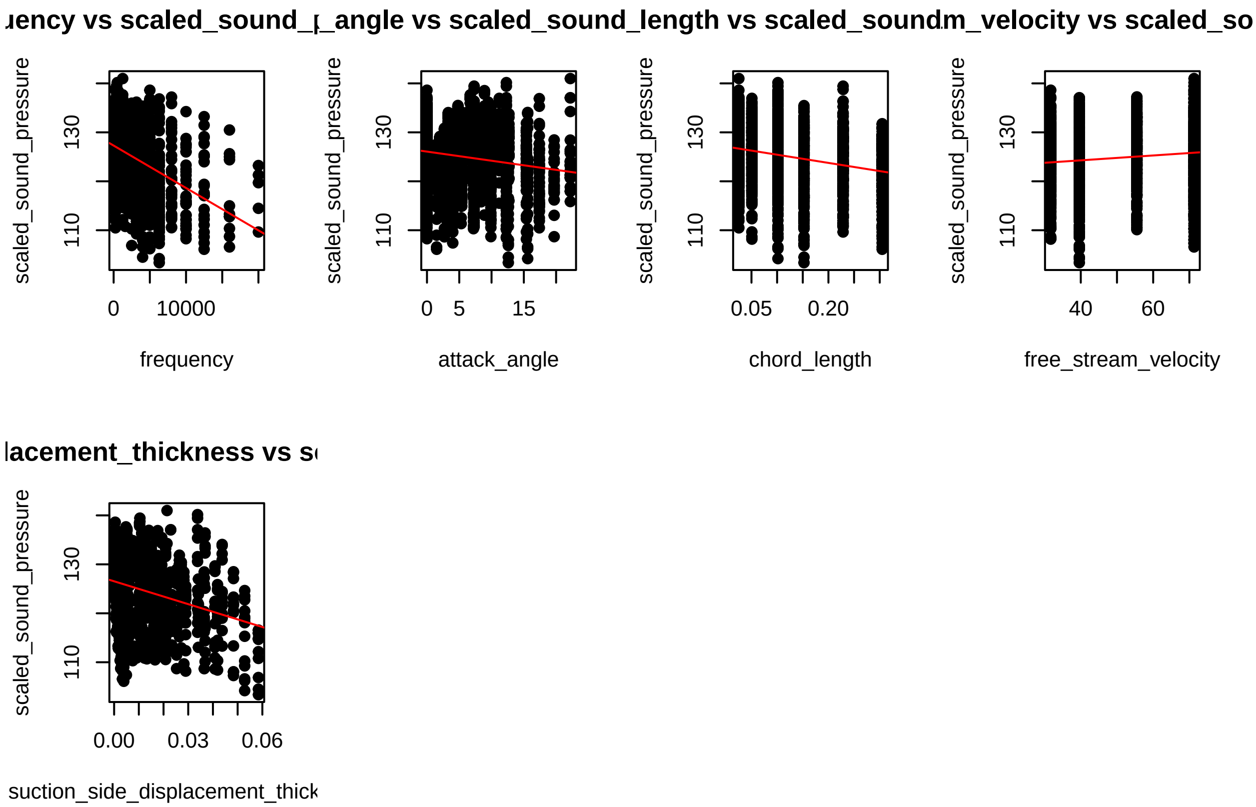
1.3.5 Check multicollinearity using VIF

Table 9: Variance Inflation Factor (VIF)

	VIF
frequency	1.14
attack_angle	3.48
chord_length	1.50
free_stream_velocity	1.05
suction_side_displacement_thickness	2.58

Every VIF is comfortably below the conventional threshold of 5, with attack_angle the highest at 3.48. This confirms the correlation-matrix finding above: the airfoil predictors share only mild, tolerable overlap, so no predictor needs to be dropped for redundancy, and OLS in Problem 2 is not expected to suffer severe coefficient instability from multicollinearity. The moderate `attack_angle` / `suction_side_displacement_thickness` correlation is still the largest source of shared variance in the data and remains worth tracking through ridge and lasso, but it falls well short of the kind of collinearity (VIF $\gg$ 5) that would make variable selection a necessity rather than a convenience.

1.3.6 Predictor-response plots



The fitted red lines are visibly weak for every panel - the same $|r| < 0.4$ ceiling seen in the correlation table above - but the point clouds are not simply noise around those lines. The `frequency` panel in particular shows the response bending down over its range rather than following the straight line, echoing the linear-vs-quadratic R^2 comparison from 1B: the straight-line fit misses curvature that a richer model could capture. The `chord_length` and `free_stream_velocity` panels show their points arranged in 6 and 4 vertical strips (their fixed experimental levels), each strip itself spanning a wide range of the response - visual confirmation that these two variables alone say little about the outcome, and any predictive power must come from combining all five predictors together, which is exactly what Problem 2's models are built to do.

1.3.7 Question

Taken together, the checks in 1A-1C describe a clean but weak-signal regression problem, and that combination shapes what the rest of Problem 1 needs to prioritise.

Data quality is not the challenge here. There are no missing values, no duplicated rows, and the "outliers" flagged by the $1.5\times$IQR rule are legitimate wind-tunnel test conditions from a controlled experiment, not data-entry errors - so cleaning required no row deletions or imputation, only the decision (1B) to retain every training row and to skip a log transform after checking that it does not help the linear fit.

The predictors are almost uncorrelated with each other. All VIFs sit well under 5, and the one moderately correlated pair (`attack_angle` / `suction_side_displacement_thickness`, $r \approx 0.76$) falls short of severe multicollinearity. This means ridge and lasso in Problem 2 are not being asked to rescue an ill-conditioned design matrix the way they would with, say, several near-duplicate predictors; their main job here is tuning the bias-variance trade-off and (for lasso) testing whether any predictor can be dropped outright, rather than resolving instability.

The predictors are individually weak but the relationships are nonlinear. Every predictor-response correlation is under 0.4 in magnitude, and no single predictor separates high- from low-noise conditions on its own (the `chord_length` and `free_stream_velocity` panels above make this explicit, with each fixed level covering a wide spread of the response). At the same time, `frequency` fits a quadratic term noticeably better than a straight line (1B), and three of the five predictors are effectively small sets of experimental design levels (1A) rather than smooth continuous measurements. Both point to the same conclusion the assignment brief anticipates for this dataset: the frequency-response relationship, and likely others, are genuinely nonlinear, so a **linear baseline (OLS) is expected to leave a meaningful amount of predictable structure on the table**, and its residuals in Problem 2 should be inspected for exactly this kind of unmodelled curvature rather than assumed to be pure noise.

2 OLS, Ridge, and Lasso

2.1 OLS baseline

```r
# Convert x_train (matrix) and y_train into a data frame for lm()
train_data <- data.frame(y = y_train, x_train)

# Fit ordinary least squares on the standardized training set
ols_fit <- lm(y ~ ., data = train_data)
```

Table 10: OLS coefficient estimates on standardized training data.

term	estimate	std.error	statistic	p.value
(Intercept)	124.8073	0.1380	904.5357	0
frequency	-4.0882	0.1476	-27.7043	0
attack_angle	-2.6498	0.2574	-10.2926	0
chord_length	-3.2617	0.1691	-19.2885	0
free_stream_velocity	1.5428	0.1411	10.9313	0
suction_side_displacement_thickness	-1.7413	0.2215	-7.8609	0

All estimates sit on the standardized scale fixed in 1B, so their magnitudes are directly comparable across rows and the intercept is the training mean of `scaled_sound_pressure`. A slope whose sign contradicts the predictor's marginal correlation with the response (1C) is the first symptom of collinearity, examined below.

```r
# In-sample fit quality
ols_summary <- summary(ols_fit)

ols_summary
```

```
##
## Call:
## lm(formula = y ~ ., data = train_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.5492  -2.8475  -0.1915   3.1609  16.4474
##
## Coefficients:
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                        124.8073     0.1380 904.536  < 2e-16 ***
## frequency                           -4.0882     0.1476 -27.704  < 2e-16 ***
## attack_angle                        -2.6498     0.2574 -10.293  < 2e-16 ***
## chord_length                        -3.2617     0.1691 -19.289  < 2e-16 ***
## free_stream_velocity                 1.5428     0.1411  10.931  < 2e-16 ***
## suction_side_displacement_thickness  -1.7413     0.2215  -7.861 8.46e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.784 on 1196 degrees of freedom
## Multiple R-squared:  0.5138, Adjusted R-squared:  0.5118
## F-statistic: 252.8 on 5 and 1196 DF,  p-value: < 2.2e-16
```

```r
# Training error (optimistic: the model has already seen these rows)
ols_train_pred    <- predict(ols_fit, newdata = train_data)
ols_train_metrics <- score_regression(y = y_train, prediction = ols_train_pred)

# Five-fold CV using the SAME foldid later passed to ridge and lasso.
# lm() has no built-in CV, so refit on each four-fold training set and
# score the held-out fold.

K <- max(foldid)
cv_mse_folds <- numeric(K)
for (k in seq_len(K)) {
  idx_train_k <- which(foldid != k)
  idx_val_k   <- which(foldid == k)
  fold_data   <- data.frame(y = y_train[idx_train_k], x_train[idx_train_k, , drop
↪  = FALSE])
  fold_fit    <- lm(y ~ ., data = fold_data)
  fold_pred   <- predict(fold_fit,
                         newdata = data.frame(x_train[idx_val_k, , drop =
                          ↪ FALSE]))
```

```
  cv_mse_folds[k] <- mean((y_train[idx_val_k] - fold_pred)^2)
}
ols_cv_mse  <- mean(cv_mse_folds)
ols_cv_rmse <- sqrt(ols_cv_mse)

# Standard error of the CV-MSE across folds. cv.glmnet reports the same quantity
# as `cvsd` for ridge and lasso, so 2D can compare all three on equal footing.
ols_cv_se <- sd(cv_mse_folds) / sqrt(K)
```

Table 11: OLS in-sample fit and five-fold cross-validation error on the shared folds.

Quantity	Value
R-squared	0.5138
Adjusted R-squared	0.5118
Training RMSE	4.7718
Training MAE	3.7081
5-fold CV RMSE	4.7988
5-fold CV MSE	23.0281
5-fold CV SE (of MSE)	1.0352

The table gathers the in-sample fit and the five-fold cross-validation error on the shared folds. The training rows are optimistic because the model was scored on the same rows it was fitted on, whereas the CV rows estimate error on unseen rows. The final row - the standard error of the CV-MSE across folds - is the quantity `cv.glmnet` reports as `cvsd` for ridge and lasso, so 2D can compare all three methods with error bars rather than bare point estimates.

Model summary. The OLS model is fitted on the 1202 standardized training observations using all 5 candidate predictors, with no penalty and no variable selection. It is therefore the unregularized baseline against which ridge and lasso are judged in 2B–2C.

Coefficient interpretation. All five predictors are highly significant ($p < 0.001$ for every one, table above), so the ranking below is by coefficient magnitude rather than by which predictors "matter".

- `frequency` carries the largest standardized coefficient ($\hat{\beta} \approx -4.09$), confirming it as the strongest single linear driver of the response, consistent with it also having the strongest marginal correlation in 1C.
- `chord_length` ($\hat{\beta} \approx -3.26$) and `attack_angle` ($\hat{\beta} \approx -2.65$) are the next largest contributors, both negative like `frequency`.
- `free_stream_velocity` ($\hat{\beta} \approx 1.54$) is the only predictor with a **positive** coefficient: higher free-stream velocity is associated with a higher (louder) scaled sound pressure once the other four predictors are held fixed.
- `suction_side_displacement_thickness` ($\hat{\beta} \approx -1.74$) has the smallest coefficient magnitude of the five, despite being the second-strongest predictor on its own in 1C ($r \approx -0.3$)
  - the gap between its marginal and partial effect is explained below.

Multicollinearity check. No coefficient here flips sign relative to its marginal correlation with the response, so there is no `lcp`-style anomaly to report. The largest variance inflation factor is 3.48 (attack_angle), comfortably below the conventional threshold of 5, consistent with the VIF table in 1C. The one predictor pair sharing meaningful variance is `attack_angle` and `suction_side_displacement_thickness` ($r \approx 0.76$, the strongest predictor-predictor correlation found in 1C), and this is visible in the coefficient ranking above: `suction_side_displacement_thickness` correlates with the response almost as strongly as `chord_length` does on its own (1C), yet ends up with the smallest partial coefficient of the five, consistent with part of its explanatory power being absorbed by the correlated `attack_angle` once both enter the model jointly. Unlike the `lcp`-style sign flip this moderate sharing produces no sign reversal or loss of significance here, but it is the same underlying mechanism, and 2B-2C test whether ridge and lasso redistribute this pair's coefficients further as the penalty increases.

```r
ols_std_resid <- rstandard(ols_fit)

qq <- qqnorm(ols_std_resid, plot.it = FALSE)

plot(qq$x, qq$y,
     type = "n",
     main = "Normal Q-Q - OLS standardized residuals",
     xlab = "Theoretical quantiles",
     ylab = "Standardized residuals")

grid(col = "grey88", lty = 1, lwd = 0.6)

abline(0, 1, col = "firebrick", lwd = 1.5, lty = 2)

points(qq$x, qq$y,
       pch = 19,
       col = "steelblue",
       cex = 0.8)

worst_resid <- order(abs(ols_std_resid), decreasing = TRUE)[1:3]

text(
  qq$x[worst_resid],
  qq$y[worst_resid],
  labels = train_id[worst_resid],
  pos = ifelse(qq$x[worst_resid] > 0, 2, 4),
  cex = 0.75,
  col = "firebrick"
)
```

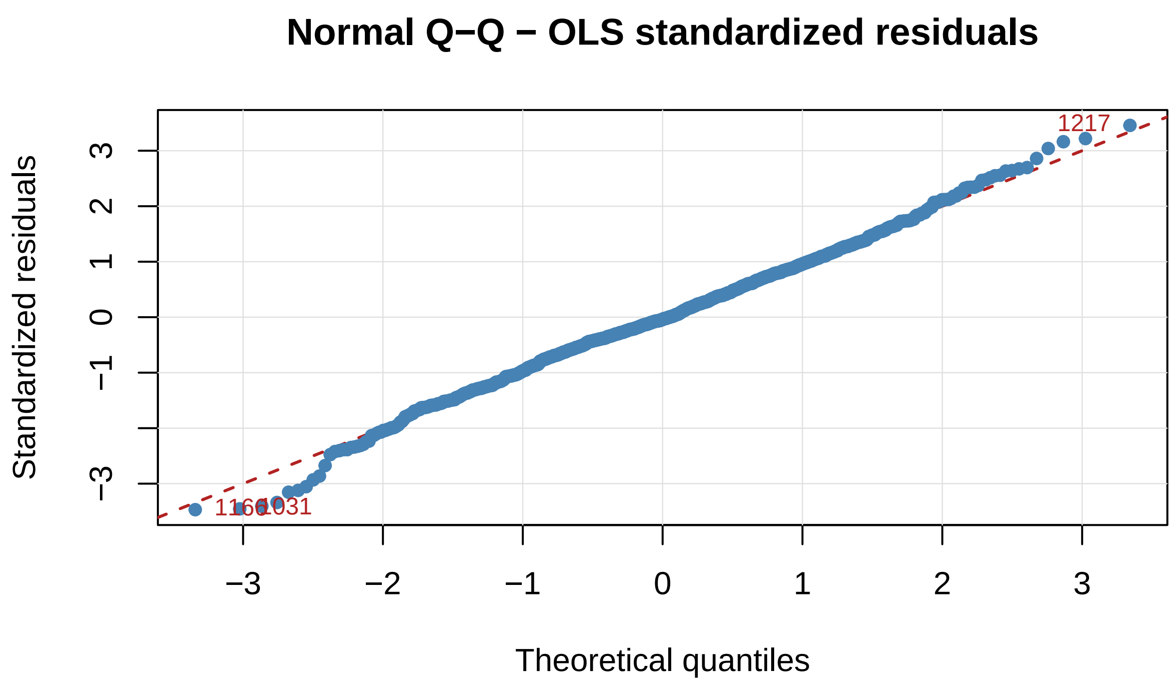


Figure 1: Normal Q-Q plot of the OLS standardized residuals against the 45-degree reference line. Points lie on the line when the errors are normal; the three largest departures are labelled with their row number in the original airfoil_self_noise data.

```
shapiro_ols    <- shapiro.test(ols_std_resid)
big_resid_rows <- train_id[which(abs(ols_std_resid) > 2)]
max_qq_gap     <- max(abs(qq$y - qq$x))
```

Residual normality. The p-values quoted above are t-tests, and they lean on the errors being approximately normal, so the assumption is checked rather than taken on trust. Each point in the Q-Q plot is one wind-tunnel measurement: its standardized residual against the value a normal sample of this size would be expected to produce at that rank. Visually the points sit close to the 45-degree line - no residual departs from it by more than 0.58 standard deviations - but with 1202 training rows a Shapiro-Wilk test has enough power to detect even small departures, and here it does ($W = 0.996$, $p = 0.0025$): formally, normality is rejected at the 5% level. W itself is close to 1, so the rejection reflects a large sample picking up a mild, not a severe, departure. 63 residuals fall beyond two standard deviations, against roughly 60 expected by chance in a sample this size - a noticeable excess, consistent with slightly heavier tails than a normal distribution rather than a handful of extreme outliers.

Two consequences follow. The coefficient p-values in the table above should be read as approximately, rather than exactly, valid, and because the tails are a little heavier than normal, RMSE - which squares errors and so weights the worst misses more heavily - is a somewhat more outlier-sensitive summary of error here than it would be under strict normality; MAE in the same table is the more robust companion figure for that reason.

Cross-validation error. The five-fold CV RMSE (4.799) is computed using the same `foldid` assignment that will be passed to `cv.glmnet` for ridge and lasso, ensuring a like-for-like comparison. Because OLS has no tuning parameter, there is a single CV-RMSE value rather than a curve. It exceeds the training RMSE (4.772) by only about 0.027 on the scaled-sound-pressure (dB) scale. This modest optimism gap reflects mild in-sample overfitting: it is small enough to suggest the 5-predictor model already generalizes reasonably well on the 1202-row training sample, and 2B–2C will test whether shrinkage narrows the gap further.

That single CV number is itself an average of 5 noisy pieces. The per-fold MSE ranges from 19.7 to 25.33 around a mean of 23.03, giving a standard error of 1.035. With about 240 observations in each held-out fold this fixes the resolution of every comparison that follows: two methods whose CV-MSE differ by less than roughly one standard error are not distinguishable on this evidence, and 2D must weigh the ridge and lasso results against that yardstick rather than declare a winner on a bare point estimate.

```
# Residual / influence diagnostic: Cook's distance
cooks_d   <- cooks.distance(ols_fit)
n_train   <- nrow(train_data)
threshold <- 4 / n_train

# Flagged points are labelled by their ORIGINAL row (via train_id) so a flagged
# observation can be traced back to the raw data.
flagged_pos <- which(cooks_d > threshold)
flagged_row <- train_id[flagged_pos]

# Headroom above the tallest bar, otherwise its label is clipped off the panel.
plot(cooks_d,
     type = "h", lwd = 1.5, col = "steelblue",
     ylim = c(0, max(cooks_d) * 1.30),
     ylab = "Cook's distance", xlab = "Training observation index",
     main = "Cook's Distance - OLS Baseline")
abline(h = threshold, lty = 2, col = "firebrick", lwd = 1.5)
if (length(flagged_pos)) {
  # Labels are rotated: adjacent flagged bars would otherwise overprint each
  ↪ other.
  text(flagged_pos, cooks_d[flagged_pos] + max(cooks_d) * 0.03,
       labels = flagged_row, srt = 90, adj = 0,
       cex = 0.45, col = "firebrick", xpd = TRUE)
}
```

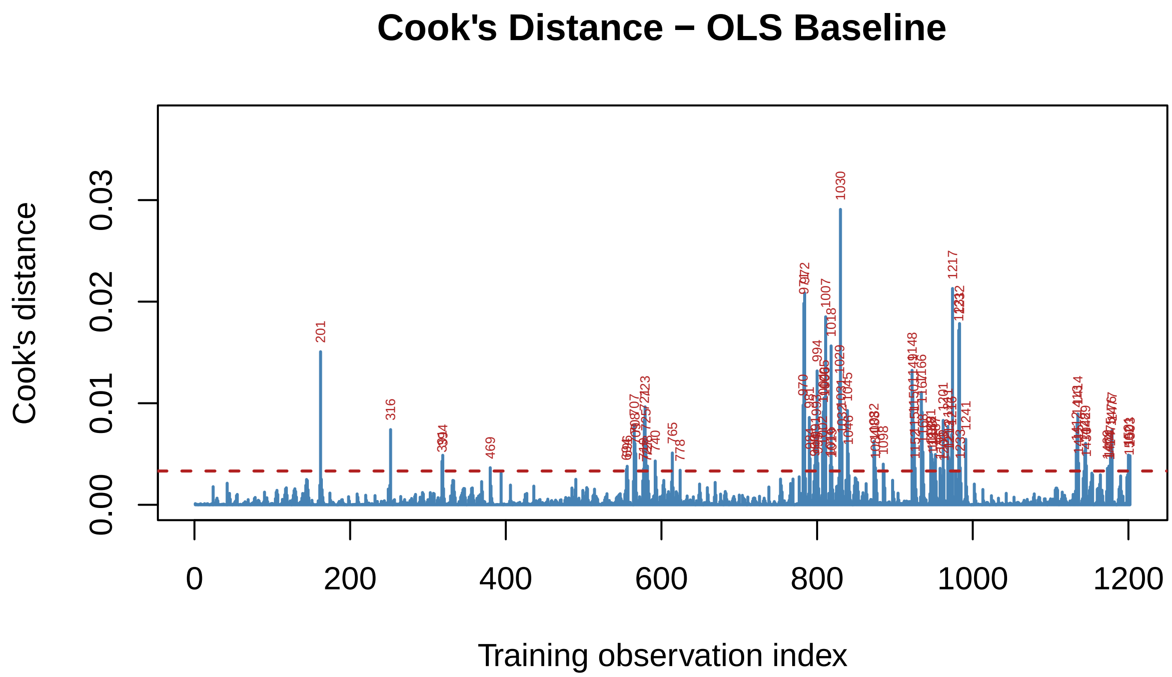


Figure 2: Cook's distance for the OLS fit. Bars above the dashed 4/n threshold are flagged as influential; each flagged bar is labelled with its row number in the original airfoil_self_noise data, not its training position.

```r
# Locate the most influential row, and two contrasting rows that separate
# leverage from residual size, all reported by their original data-row number.
n_flagged   <- length(flagged_pos)
top_pos     <- flagged_pos[which.max(cooks_d[flagged_pos])]
top_row     <- train_id[top_pos]

# Leverage separates the two ingredients Cook's distance mixes together.
ols_leverage <- hatvalues(ols_fit)
avg_leverage <- length(coef(ols_fit)) / n_train

# High-leverage / low-residual example: the training row with the largest
# leverage, which turns out to sit at the maximum observed frequency (1A/1C).
hilev_pos <- which.max(ols_leverage)
hilev_row <- train_id[hilev_pos]
hilev_is_flagged <- hilev_pos %in% flagged_pos

# Low-leverage / high-residual example: the training row with the largest
# standardized residual, an ordinary point in predictor space that the model
# simply predicts badly.
hires_pos <- which.max(abs(ols_std_resid))
hires_row <- train_id[hires_pos]
```

Influence diagnostic. Cook's distance asks how far the whole fitted coefficient vector would move if one observation were deleted, so it combines a large residual and high leverage into a single number. Against the conventional threshold $4/n \approx 0.0033$, 88 of the 1202 training rows are flagged - a large count relative to 1202, which is typical of this threshold on a big sample rather than a sign of 88 genuinely extreme rows. The single most influential row is 1030 (Cook's $D \approx 0.029$), which combines both ingredients at once: leverage 5.4 times the training average and a standardized residual of 2.51.

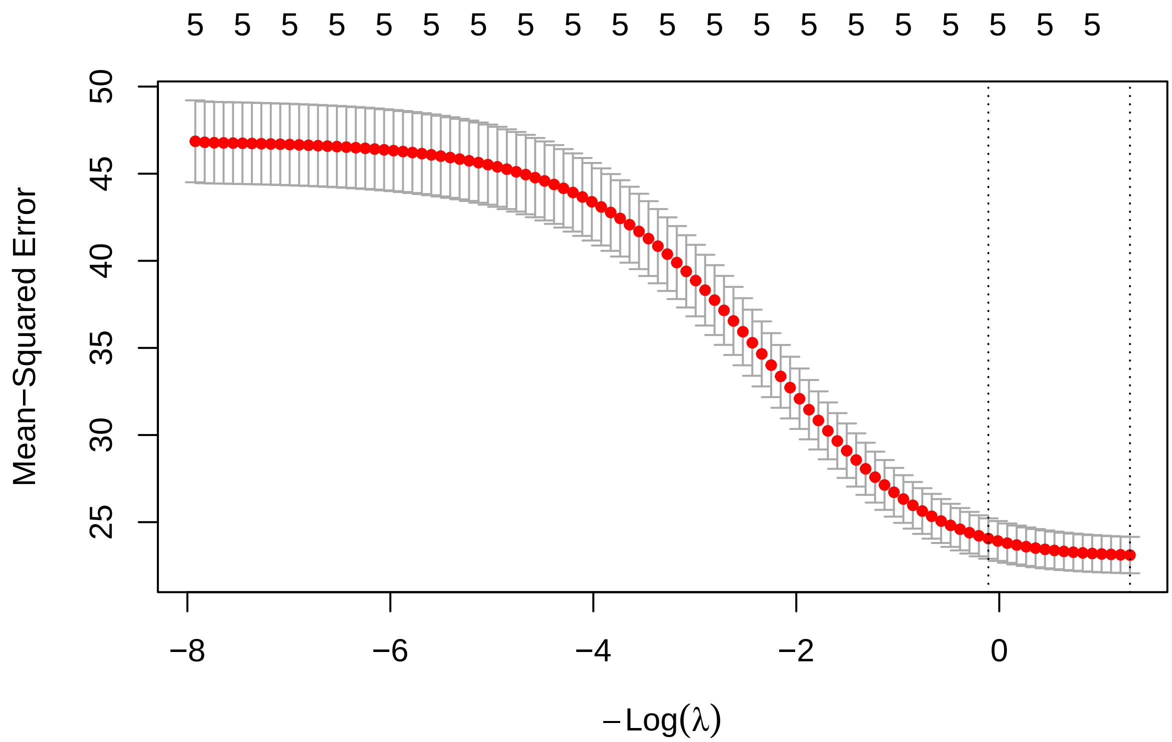
Two other flagged rows separate those ingredients cleanly and connect back to 1A/1C. Row 316 has the single largest leverage in the training set (5.7 times the average) because it sits at `frequency` = 20000 Hz, the maximum value in the dataset and the extreme right-hand tail flagged as a legitimate experimental condition rather than an error in 1A-1B. Despite that extreme position in predictor space, its standardized residual is only 1.23 - comfortably inside the normal range, so the model predicts it well. Row 1166 is the mirror image: an unremarkable position in predictor space (leverage close to the training average) but the largest standardized residual of any training row (-3.47) - a case the linear fit simply misses, not one it is destabilized by.

This confirms the two ingredients Cook's distance mixes are genuinely distinct here, and that no single kind of anomaly dominates: some rows are influential because they are extreme in predictor space, others because they are badly predicted, and the worst case (row 1030) is a row where both happen at once. All flagged observations are retained rather than removed - they are valid wind-tunnel measurements, consistent with the outlier decisions already made in 1B - and their leverage is instead absorbed by the shrinkage introduced in 2B–2C, which is one of the concrete things the regularized fits are expected to improve, particularly for the correlated `attack_angle` / `suction_side_displacement_thickness` pair whose estimates are the most sensitive to exactly this kind of point.

2.2 Ridge regression

2.2.1 Cross-validation curve:

```
ridge <- analysis_model(x = x_train, y = y_train, alpha=0, foldid = foldid)
ridge_cvsd_min <- ridge$model$cvsd[ridge$model$lambda == ridge$model$lambda.min]
```



The mean cross-validation MSE decreases as $-Log(\lambda)$ increases (equivalently, as λ decreases), suggesting that reducing the regularization strength improves predictive performance. The left dashed line indicates λ_{1se}, which selects a more regularized model with prediction error within one standard error of the minimum, while the right dashed line indicates λ_{min}, which yields the lowest cross-validation error.

2.2.2 Lambda min:

Metric	Value
Lambda min	0.2758
MSE	23.1116
Condition number	5.3423

With $\lambda_{min} = 0.2758$, the 5-fold cross-validation MSE reaches its minimum value of $MSE_{min} = 23.1116$, indicating the best predictive performance among the candidate values of λ. The corresponding condition number, 5.34, is already a substantial improvement over the unpenalized Gram matrix's 12.3 (1C/3C) - even this mildly-collinear dataset (max VIF 3.48, 2A) benefits measurably from even a small ridge penalty.

2.2.3 Lambda 1se:

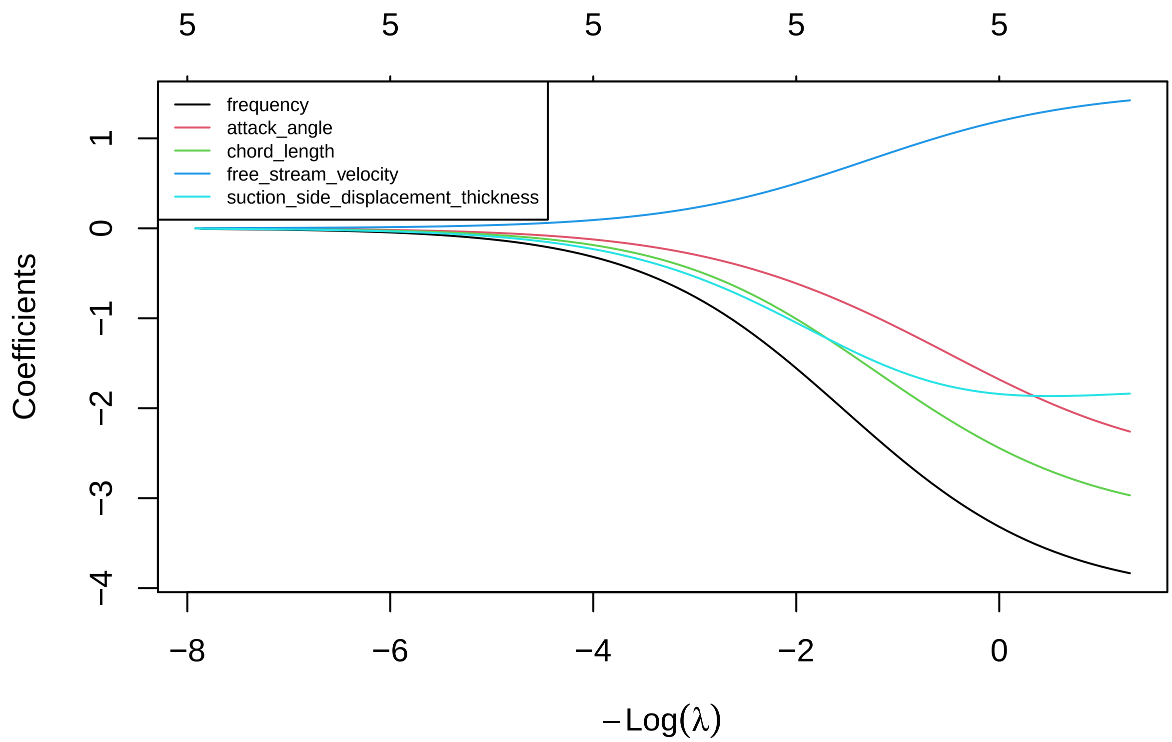
Metric	Value
Lambda 1se	1.1136
MSE	24.0592
Condition number	2.5129

Although the one-standard-error rule selected a larger penalty $\lambda_{1se} = 1.1136$, the condition number decreased further from 5.34 to 2.51. In contrast, the cross-validation MSE increased from 23.112 to 24.059, a gap of 0.948 that is smaller than ridge's own cross-validation standard error at λ_{min} (1.044, the `cvsd` `cv.glmnet` reports, mirroring the per-fold SE computed for OLS in 2A) - the two penalties are not statistically distinguishable on this evidence, so the extra stability from λ_{1se} is close to free in prediction-error terms.

2.2.4 Coefficients:

	OLS	Lambda_Min	Lambda_Max
(Intercept)	124.807270	124.807270	124.807270
frequency	-4.088176	-3.834522	-3.250734
attack_angle	-2.649795	-2.260002	-1.619908
chord_length	-3.261692	-2.968539	-2.379795
free_stream_velocity	1.542842	1.422378	1.162427
suction_side_displacement_thickness	-1.741310	-1.837350	-1.829411

Compared with the OLS model, Ridge regression shrinks most regression coefficients toward zero, reducing their magnitudes through the regularization penalty, while retaining all five predictors in the model. `frequency`, `attack_angle`, `chord_length`, and `free_stream_velocity` all shrink exactly as expected, moving closer to zero as the penalty strengthens. `suction_side_displacement_thickness` is the one exception: its magnitude at λ_{min} (-1.837) is slightly *larger* than its OLS estimate (-1.741), moving further from zero rather than shrinking. This is the same mechanism flagged in 2A: `suction_side_displacement_thickness` shares variance with the correlated `attack_angle` ($r \approx 0.76$), and as Ridge shrinks `attack_angle`'s coefficient it redistributes some of that shared explanatory power back onto `suction_side_displacement_thickness`. Unlike a severe collinearity case, no sign changes anywhere in the table - the exception is a small redistribution, not an instability.



As λ increases, all five coefficients are progressively shrunk toward zero, demonstrating the regularization effect of Ridge regression; as λ decreases, they move back toward their OLS estimates. `frequency` remains the largest-magnitude coefficient across the range spanned by λ_{min} and λ_{1se} (-3.83 to -3.25), consistent with it being the strongest predictor throughout 1C-2A. `free_stream_velocity` is the only coefficient that stays positive along the entire path, mirroring the sign pattern already seen in OLS. No coefficient crosses zero or reverses sign between λ_{min} and λ_{1se} - the ridge penalty here is redistributing magnitude among correlated predictors (as seen above for `suction_side_displacement_thickness`), not fighting an unstable or sign-flipping fit.

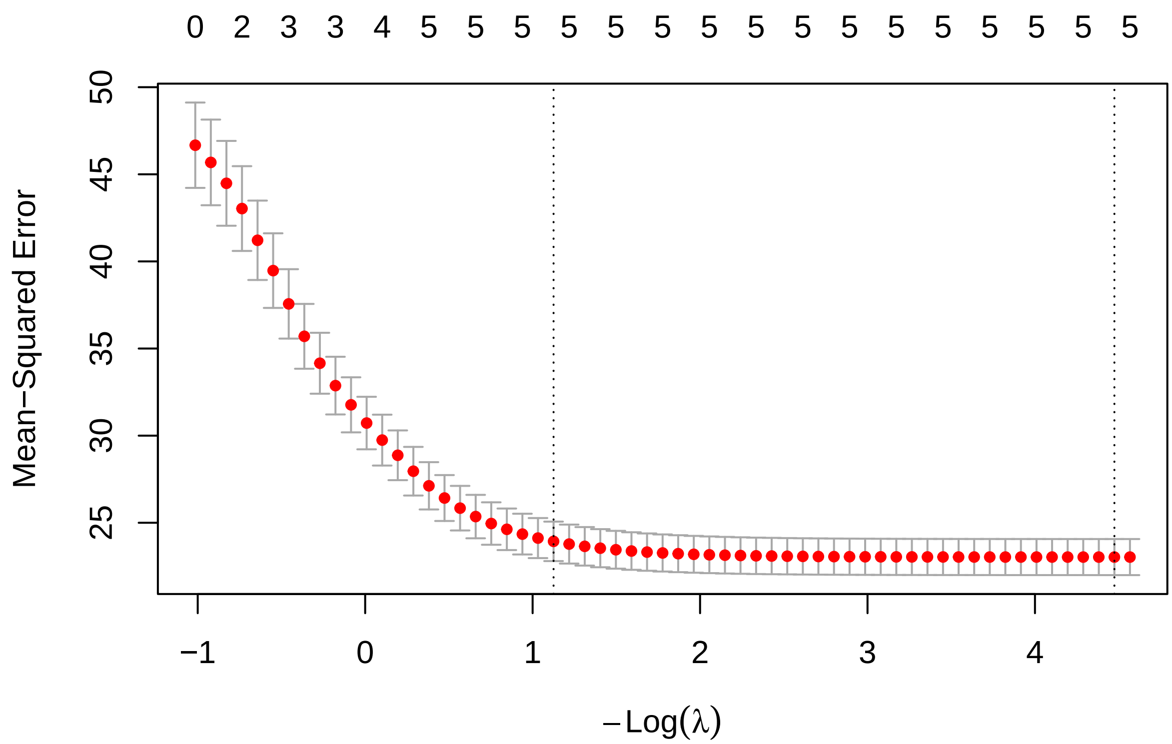
To address multicollinearity, Ridge impose an L_2 penalty on the regression coefficients. Rather than assigning a large coefficient to one correlated predictor and a small coefficient to another, Ridge shrinks the coefficients simultaneously and redistributes their effects among the correlated variables. Consequently, the coefficient estimates become more stable and less sensitive to small changes in the data.

2.3 Lasso regression

2.3.1 Model fitting and cross-validation

```
lasso_cv <- analysis_model(
    x = x_train,
    y = y_train,
```

```
    alpha = 1,
    foldid = foldid
)
```



Based on the 5-fold cross-validation curve generated for the Lasso regression model ($\alpha = 1$), we observe the relationship between the tuning parameter λ and the Mean-Squared Error (MSE). The top axis indicates the number of active (non-zero) predictors retained in the model at each level of penalization.

- The Cross-Validation Curve: The red dots represent the average CV-MSE across the 5 folds for each λ value, with the gray error bars denoting one standard error above and below the mean. As $-\log(\lambda)$ increases (moving from left to right, meaning the penalty weakens), the MSE initially drops, reaches a minimum, and then slightly stabilizes as the model approaches the unpenalized OLS limit.
- Optimal λ (λ_{min}): The right vertical dashed line marks `lambda.min` ($-\log(\lambda) \approx 4.47$), the value that strictly minimizes the cross-validation MSE. At this optimal point for prediction accuracy the Lasso penalty retains all 5 predictors.
- Parsimonious λ (λ_{1se}): The left vertical dashed line marks `lambda.1se` ($-\log(\lambda) \approx 1.13$), selected by the one-standard-error rule. This value applies a stronger penalty to produce a more conservative model while keeping the error within one standard deviation of the absolute minimum MSE; whether it also produces a *sparser* model, in the sense of dropping predictors, is checked directly below rather than assumed.

2.3.2 Analysis of λ_{min}

Metric	Value
Lambda min	0.0114
MSE	23.0305
Condition number	11.6000

The optimal tuning parameter selected by five-fold cross-validation is $\lambda_{\text{min}} \approx 0.0114$. This value strictly minimizes the cross-validation Mean Squared Error to 23.03, essentially matching OLS's own CV-MSE (23.028, 2A) - unsurprising given how small λ_{min} is. The condition number at this threshold, 11.6, is a modest improvement over the unpenalized Gram matrix (12.3, 1C/3C).

At this level of regularization, the Lasso model retains all 5 predictors - `frequency`, `attack_angle`, `chord_length`, `free_stream_velocity`, and `suction_side_displacement_thickness` - none is shrunk exactly to zero. This is consistent with the OLS and VIF results in 2A: every predictor was already highly significant with only mild shared variance, so there is no clearly redundant predictor for Lasso's L_1 penalty to remove at this tuning value.

The retained coefficients preserve the same ranking seen in OLS and ridge: `frequency` has the largest magnitude (-4.065), followed by `chord_length` (-3.232) and `attack_angle` (-2.61), with `free_stream_velocity` (1.525) the only positive coefficient and `suction_side_displacement_thickness` (-1.748) the smallest in magnitude.

Overall, λ_{min} produces a model whose cross-validation error is essentially indistinguishable from OLS's while retaining every predictor: on this dataset, Lasso's contribution at λ_{min} is confirming that all five predictors carry real signal, rather than selecting a subset of them.

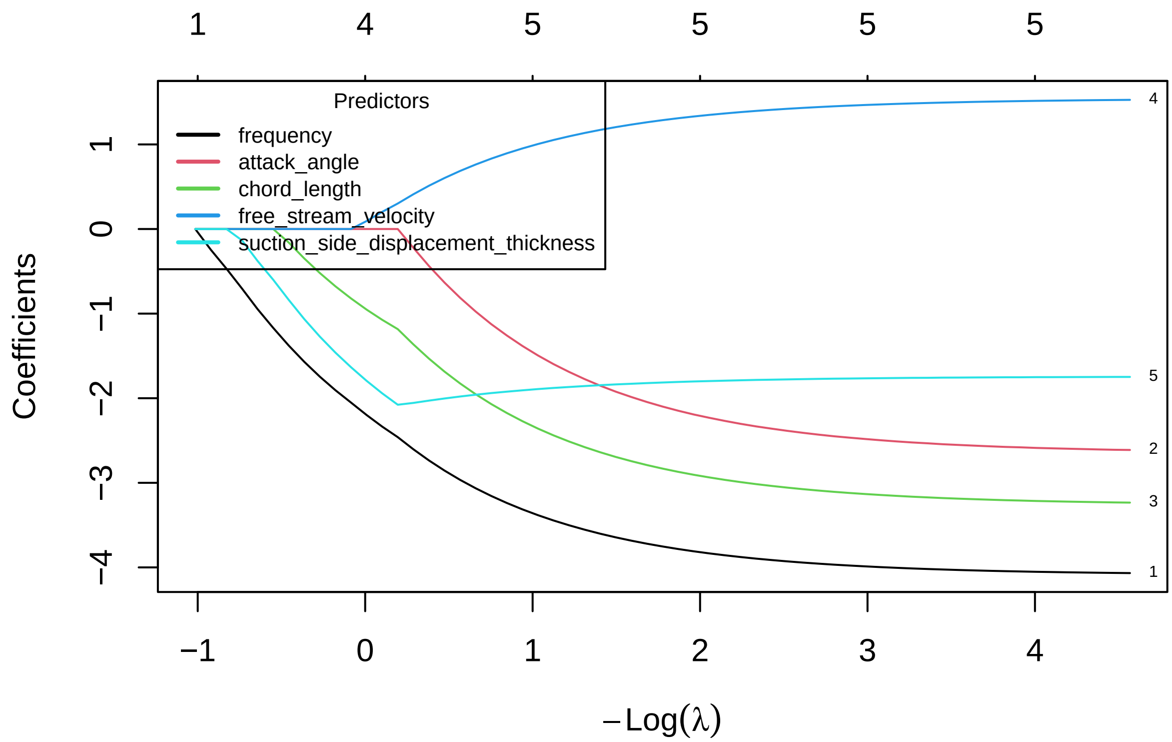
2.3.3 Analysis of λ_{1se}

Metric	Value
Lambda 1se	0.3246
MSE	23.9327
Condition number	4.9159

The one-standard-error rule selects $\lambda_{\text{1se}} \approx 0.3246$, which applies a stronger regularization penalty than λ_{min}. The corresponding five-fold cross-validation MSE is 23.933. Although this is higher than the minimum achieved by λ_{min}, it remains within one standard error of that minimum by construction. The stronger penalty reduces the condition number to 4.92, a substantial further improvement in conditioning.

Under λ_{1se}, the Lasso model still retains all 5 predictors - unlike the sparser model this rule often produces, no predictor is driven to zero even under this stronger penalty. Coefficient magnitudes shrink further relative to λ_{min} (for example `frequency` falls from -4.065 to -3.445), but the set of active predictors is unchanged. Combined with the λ_{min} result above, this indicates that on this dataset Lasso's embedded feature-selection mechanism finds nothing to prune: every predictor's contribution is large enough, relative to the cross-validated penalty, to survive even the more conservative tuning rule.

2.3.4 Coefficient Path



Overall, the coefficient path shows that as $-\log(\lambda)$ increases (equivalently, as λ decreases), the regularization penalty becomes weaker and coefficients grow in magnitude toward their OLS values. Under strong enough regularization all five paths do eventually collapse to zero at the far left of the plot, but by the range spanned by λ_{min} and λ_{1se} - the only penalty strengths this analysis tunes to - every predictor is already active, matching the retained-predictor counts reported above.

`frequency` enters first (i.e. survives down to the smallest λ) and consistently has the largest-magnitude coefficient along the path, confirming it as the most influential predictor. `chord_length` and `attack_angle` follow a similar trajectory at smaller magnitudes, and `free_stream_velocity` is the only path that stays on the positive side of zero throughout. `suction_side_displacement_thickness` tracks a comparatively flat, small-magnitude path across the region shown - the same predictor whose coefficient 2B found to move *against* the general shrinkage trend under ridge.

2.4 Controlled comparison and locked core model

	tunning_rule	lambda	cv_error	cond	coefficient_norm	num_non_zero
OLS	None	0.0000000	23.02805	12.301957	6.307595	5
Ridge	lambda.min	0.2758485	23.11155	5.342311	5.832873	5
	lambda.1se	1.1136050	24.05918	2.512911	4.853120	5

	tunning_rule	lambda	cv_error	cond	coefficient_norm	num_non_zero
Lasso	lambda.min	0.0113981	23.03046	11.599965	6.258213	5
	lambda.1se	0.3246218	23.93266	4.915946	5.001118	5

The training comparison tells a different story here than the classic "lasso selects, ridge stabilizes" narrative. All five candidates - OLS, ridge, and lasso at both tuning rules - retain all 5 predictors: on this dataset lasso performs no variable selection at either λ_{min} or λ_{1se} (2C), so it offers no interpretability advantage over ridge here. The five CV errors are also close together (23.03 to 24.06), well inside one cross-validation standard error of each other (1.04, 2A) - none of the five is distinguishable from any other on predictive accuracy alone. What *does* separate them is conditioning: ridge at λ_{min} nearly matches OLS's and lasso's CV error while roughly halving the condition number (5.34 vs OLS's 12.3), and λ_{1se} improves conditioning further for both ridge and lasso at a small additional cost in CV error. Because lasso does not sparsify this model, the usual accuracy-vs-interpretability trade-off does not apply; the trade-off that remains is prediction accuracy vs. numerical stability. On that basis, **ridge at** λ_{min} is nominated as the core candidate: it is statistically tied with OLS and lasso on training CV error while providing a meaningfully better-conditioned fit. This recommendation is based solely on the training data and will later be evaluated on the locked holdout set.

The λ_{min} and λ_{1se} selection rules address different modeling objectives. The λ_{min} rule selects the value of λ that minimizes the cross-validation error and therefore prioritizes predictive accuracy. In contrast, the λ_{1se} rule selects the largest λ whose cross-validation error is within one standard error of the minimum, favouring a simpler and more stable model with stronger regularization. These rules therefore answer different decision questions: whether the primary objective is the best possible prediction or a simpler, more stable model with comparable predictive performance. Predictive performance is assessed by the cross-validation error, whereas variable selection concerns which predictors remain in the model, and interpretability concerns how easily the resulting model can be understood. In our results this trade-off shows up in *conditioning* rather than in *sparsity*: λ_{1se} retains exactly as many predictors as λ_{min} for both ridge and lasso (2B-2C), but trades a small increase in cross-validation error for a meaningfully better-conditioned fit, illustrating a stability-vs-accuracy trade-off rather than a complexity-vs-accuracy one.

2.5 Perturbation or stability check

2.5.1 Prediction:

Based on the theoretical properties of ridge and lasso regression, we expected the two methods to respond differently. Ridge regression was expected to remain stable because the L2 penalty distributes the effect across correlated predictors by shrinking their coefficients toward each other. Consequently, only small changes in prediction error and coefficient estimates were anticipated. In contrast, lasso regression was expected to maintain similar prediction accuracy but perform variable selection by retaining only one predictor from the highly correlated pair. Therefore, we expected lasso's predictive performance to remain stable, while its selected variables could become less stable.

```r
x_train_new = data.frame(x_train)
x_train_new$frequency_new <- x_train_new$frequency + rnorm(nrow(x_train), 0, 0.1)
kable(cor(x=x_train_new$frequency, y = x_train_new$frequency_new))
```

x
0.994593

```r
ridge_new <- analysis_model(as.matrix(x_train_new), y_train, foldid=foldid, alpha
↪ = 0)
lasso_new <- analysis_model(as.matrix(x_train_new), y_train, foldid=foldid, alpha
↪ = 1)
```

	Old Ridge	New Ridge	Old Lasso	New Lasso
Lambda min	0.2758	0.2758	0.0114	0.0104
Lambda 1se	1.1136	1.3413	0.3246	0.3246
Mse min	23.1116	23.1078	23.0305	23.0312
Mse 1se	24.0592	24.0377	23.9327	23.9322
Coef min norm	5.8329	5.2342	6.2582	6.1241
Coef 1se norm	4.8531	4.2809	5.0011	5.0011
Condition number of min	5.3423	9.8134	11.6000	158.3438
Condition number of 1se	2.5129	2.8404	4.9159	8.5108
Nonzero min	5.0000	6.0000	5.0000	6.0000
Nonzero 1se	5.0000	6.0000	5.0000	5.0000

To investigate the stability of ridge and lasso regression, a perturbation experiment was conducted by adding a new predictor (`frequency_new`) that is a noisy duplicate of `frequency`, the strongest predictor identified in 2A ($r \approx 0.995$ with the original). This creates a pair of highly correlated predictors while preserving nearly the same information, on top of the already-correlated `attack_angle`/`suction_side_displacement_thickness` pair from 1C.

	Old Ridge's coef $\lambda = \lambda_{min}$	Old Ridge's coef $\lambda = \lambda_{1se}$	New Ridge's coef $\lambda = \lambda_{min}$	New Ridge's coef $\lambda = \lambda_{1se}$
(Intercept)	124.8073	124.8073	124.8057	124.8058
frequency	-3.8345	-3.2507	-2.1498	-1.7668
attack_angle	-2.2600	-1.6199	-2.2986	-1.5900
chord_length	-2.9685	-2.3798	-2.9914	-2.3024
free_stream_velocity	1.4224	1.1624	1.4315	1.1440
suction_side_displacement_thickness	-1.8373	-1.8294	-1.8276	-1.8131
frequency_new	NA	NA	-1.7760	-1.6671

```r
n <- max(nrow(lasso_cv$coef_min), nrow(lasso_new$coef_min))

df1 <- data.frame(
  Lasso_Coef_Min = as.numeric(lasso_cv$coef_min),
  Lasso_Coef_1se = as.numeric(lasso_cv$coef_1se)
)
df1 <- df1[1:n, ,drop=FALSE]
df2 <- data.frame(
  Lasso_New_Coef_Min = as.numeric(lasso_new$coef_min),
  Lasso_New_Coef_1se = as.numeric(lasso_new$coef_1se)
)
df2 <- df2[1:n, ,drop=FALSE]

df <- cbind(df1, df2, row.names = rownames(lasso_new$coef_min))
df <- round(df, 4)
names(df) <- c("Old Lasso's coef $\\lambda = \\lambda_{min}$", "Old Lasso's coef
↪  $\\lambda = \\lambda_{1se}$", "New Lasso's coef $\\lambda = \\lambda_{min}$",
↪  "New Lasso's coef $\\lambda = \\lambda_{1se}$")
knitr::kable(df)
```

	Old Lasso's coef $\lambda = \lambda_{min}$	Old Lasso's coef $\lambda = \lambda_{1se}$	New Lasso's coef $\lambda = \lambda_{min}$	New Lasso's coef $\lambda = \lambda_{1se}$
(Intercept)	124.8073	124.8073	124.8071	124.8073
frequency	-4.0654	-3.4454	-3.8418	-3.4454
attack_angle	-2.6095	-1.5974	-2.6157	-1.5973
chord_length	-3.2318	-2.4395	-3.2362	-2.4395
free_stream_velocity	1.5254	1.0527	1.5267	1.0527
suction_side_displacement_thickness	-1.7483	-1.8785	-1.7456	-1.8786
frequency_new	NA	NA	-0.2289	0.0000

Comparing the coefficient estimates before and after introducing the duplicated predictor shows two different responses at the two tuning rules. At λ_{1se}, lasso's original five coefficients are essentially unchanged (coefficient norm 5.001 before vs. 5.001 after) and `frequency_new` receives an **exact zero** coefficient ($\hat{\beta} = 0$) - the textbook lasso behaviour of picking one predictor from a near-duplicate pair and dropping the other entirely. At λ_{min}, however, the picture is more nuanced: `frequency` keeps almost all of its original coefficient (-4.065 before, -3.842 after) but `frequency_new` is *not* zeroed out, receiving a small nonzero weight of -0.229 - lasso's active-set count at λ_{min} rises from 5 to 6. So lasso's often-cited "picks one and zeros the rest" behaviour held here, but only at the more conservative λ_{1se} penalty; at λ_{min} the penalty is too weak to force the duplicate fully to zero.

Ridge's response is the mirror image at both tuning rules: rather than favouring one of the pair, it splits the original `frequency` coefficient roughly evenly between `frequency` (-2.15) and `frequency_new` (-1.776) at λ_{min}, and similarly at λ_{1se} (-1.767 and -1.667), consistent with the L_2 penalty redistributing shared explanatory power rather than selecting a winner.

2.5.2 Result.

The table above summarizes the results before and after introducing the duplicated predictor. For ridge regression, λ_{min} itself is unchanged (0.2758) and the cross-validation MSE barely moves (23.11 to 23.11), but the condition number at λ_{min} nearly doubles (5.34 to 9.81) - adding a highly correlated predictor measurably worsens conditioning even though ridge handles it without any loss of predictive accuracy. At λ_{1se} the effect is much smaller (2.51 to 2.84), because the larger penalty already dominates the added collinearity. Ridge's coefficient norm decreases slightly at both tuning rules (5.83 to 5.23 at λ_{min}) because splitting one large coefficient into two smaller, correlated ones reduces the sum of squares even though the combined effect on the fit is nearly the same. As expected for ridge, the number of nonzero coefficients simply tracks the number of predictors offered to it (5 to 6).

For lasso, predictive performance is similarly unaffected (CV-MSE 23.03 to 23.03 at λ_{min}; 23.93 to 23.93 at λ_{1se}), but the condition number tells a sharper story than ridge's. At λ_{min} the condition number jumps from 11.6 to 158.34 - far more than ridge's - because λ_{min} is small enough that the fit sits close to the ill-conditioned unpenalized regression on a near-collinear pair. At λ_{1se} the increase is much smaller (4.92 to 8.51), because the stronger penalty - and the exact zero it places on `frequency_new` - keeps the effective design well away from collinearity.

The experimental results are largely consistent with theoretical expectations, with one refinement. Ridge handled the added multicollinearity through coefficient redistribution rather than selection, maintaining stable predictive performance throughout, exactly as expected. Lasso's variable-selection behaviour, however, turned out to depend on the tuning rule: it is stable and sparse - selecting `frequency` and exactly zeroing `frequency_new` - only at the more conservative λ_{1se}, while at λ_{min} it keeps both predictors active with unequal weights, closer to ridge's redistribution behaviour than to a clean selection. This matters for 2D's earlier point: on this dataset lasso already retained every predictor before any perturbation (2C), and this experiment shows its selection mechanism is most reliable at the more regularized λ_{1se} tuning, reinforcing λ_{1se} - not λ_{min} - as the more trustworthy choice whenever lasso's sparsity is being relied upon for interpretation.

3 Experimental Design

3.1 Dataset selection

3.1.1 Data Overview

```
data <- read.csv("data/penguins_size.csv", header = TRUE)

# Treat "." as a missing value
data[data == "."] <- NA

config <- list(
  file = "penguins_size.csv",
  response = "body_mass_g",
  factors = c("species", "sex")
)
```

```r
data_raw <- data

analysis_data <- data_raw[
  ,
  c(config$response, config$factors),
  drop = FALSE
]

colSums(is.na(analysis_data))
```

```
## body_mass_g     species         sex
##           2           0          11
```

```r
analysis_data <- na.omit(analysis_data)

if (anyNA(analysis_data)) {
  stop("Missing values remain in the analysis data.")
}
```

Table 12: Data Overview Before Missing-Value Removal

Variable	Role	Type	Distinct	Missing
species	Predictor	character	3	0
island	Predictor	character	3	0
culmen_length_mm	Predictor	numeric	164	2
culmen_depth_mm	Predictor	numeric	80	2
flipper_length_mm	Predictor	integer	55	2
body_mass_g	Response	integer	94	2
sex	Predictor	character	2	11

For 2-Way ANOVA, this method requires the dataset to have at least 2 categorical variables along with a continuous response variable for the analysis.

In general, from our chosen dataset, we decide to evaluate the factors that are associated with penguin body mass. This dataset contains information about 344 penguins, including species, island, physical measurements such as culmen length, culmen depth, and flipper length, body mass, and sex. It is suitable for exploratory data analysis and 2-Way ANOVA analysis based on the biological and physical characteristics of penguins.

Specifically, the dataset contains 3 categorical variables: `species` (Adelie/Chinstrap/Gentoo), `island` (Torgersen/Biscoe/Dream), and `sex` (MALE/FEMALE). In this method, we select `species` and `sex` as the two categorical factors and `body_mass_g` as the continuous response variable. The analysis is used to test whether the two factors are associated with differences in mean body mass, as well as whether there is an interaction between the two factors.

3.1.2 Research Questions & Hypotheses

We investigate whether penguin species and sex are associated with differences in mean body mass, and whether there is an interaction between the two factors.

We follow the 2-way ANOVA model:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

with $\epsilon_{ijk} \sim N(0, \sigma^2)$.

- μ is the overall mean.
- α_i is the effect of the i-th level of Factor A, `species`.
- β_j is the effect of the j-th level of Factor B, `sex`.
- $(\alpha\beta)_{ij}$ is the interaction effect between Factor A and Factor B.
- ϵ_{ijk} is the random error term.

3.1.2.1 Factor A: Penguin Species **Research Question:**
Is penguin species associated with differences in mean body mass?

$$\begin{cases} H_{0a}: \alpha_1 = \alpha_2 = \alpha_3 = 0 \\ H_{1a}: \text{Not all } \alpha_i \text{ are equal to 0.} \end{cases}$$

3.1.2.2 Factor B: Penguin Sex **Research Question:**
Is the sex of penguin associated with differences in mean body mass?

$$\begin{cases} H_{0b}: \beta_1 = \beta_2 = 0 \\ H_{1b}: \text{Not all } \beta_j \text{ are equal to 0.} \end{cases}$$

3.1.2.3 Interaction Effect **Research Question:**
Does the association between sex and penguin body mass depend on the penguin species?

$$\begin{cases} H_{0ab}: (\alpha\beta)_{ij} = 0 \quad \text{for all } i, j \\ H_{1ab}: \text{Not all } (\alpha\beta)_{ij} \text{ are equal to 0.} \end{cases}$$

3.1.3 Descriptive Statistics

Table 13: Overall Summary Statistics of Penguin Body Mass

N	Mean	Median	Variance	SD	Min	Max
333	4207.06	4050	648372.5	805.22	2700	6300

3.1.3.1 Data Summary Statistics

The dataset contains 333 observations of penguin body mass after removing observations with missing values. The mean body mass is 4207.06 g, with a median of 4050 g, indicating that the body masses are centered around approximately 4.0 to 4.2 kg. The standard deviation is 805.22 g, showing moderate variability in body mass, while the observed values range from 2700 g to 6300 g. The mean being slightly higher than the median suggests a mild right skew in the overall response distribution.

Table 14: Summary Statistics by Treatment Combination

Species	Sex	N	Mean	Median	Variance	SD	Min	Max
Adelie	FEMALE	73	3368.84	3400	72565.64	269.38	2850	3900
Adelie	MALE	73	4043.49	4000	120278.25	346.81	3325	4775
Chinstrap	FEMALE	34	3527.21	3550	81415.44	285.33	2700	4150
Chinstrap	MALE	34	3938.97	3950	131143.61	362.14	3250	4800
Gentoo	FEMALE	58	4679.74	4700	79286.34	281.58	3950	5200
Gentoo	MALE	61	5484.84	5500	98068.31	313.16	4750	6300

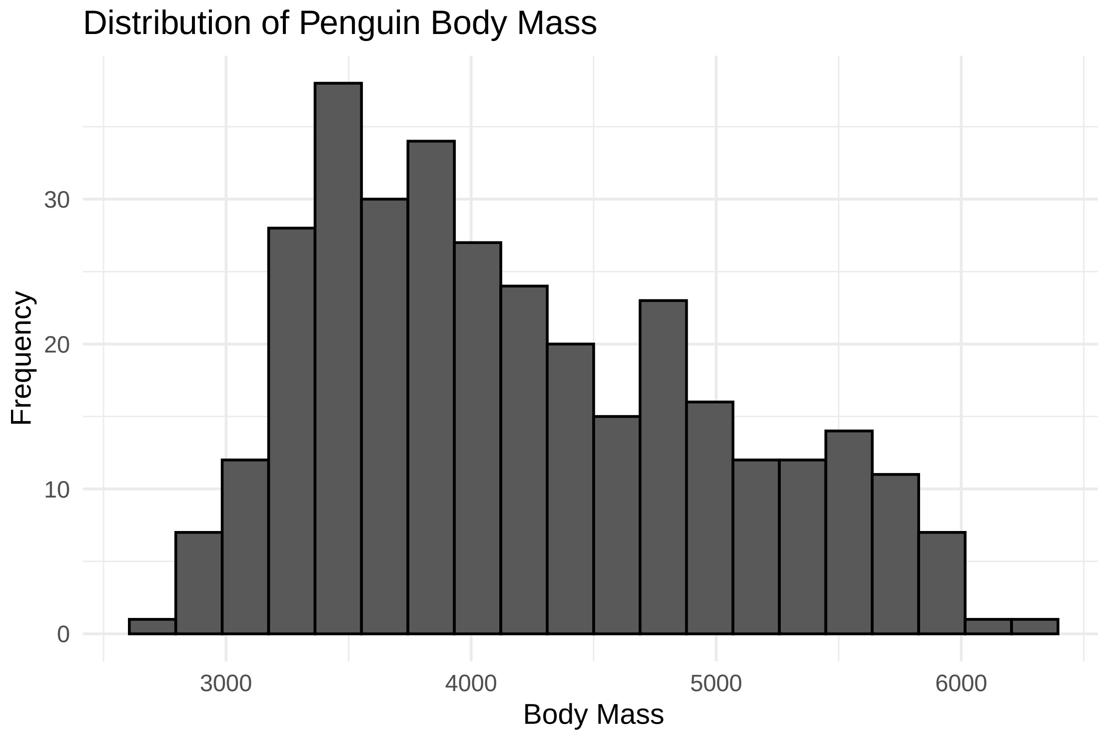
The dataset contains three levels of species and two levels of sex. Male penguins generally have higher mean body mass than female penguins across all three species.

The difference in mean body mass between males and females varies across species. The difference is approximately 674.65 g for Adelie penguins, 411.76 g for Chinstrap penguins, and 805.10 g for Gentoo penguins. This variation provides descriptive evidence of a potential interaction between species and sex.

Among the six treatment combinations, the Gentoo male group has the highest mean body mass (5484.84 g), while the Adelie female group has the lowest mean body mass (3368.84 g). The group variances range from 72,565.64 to 120,278.25, with the Adelie male group having the largest variance. Therefore, the homogeneity of variance assumption should be checked before conducting the two-way ANOVA.

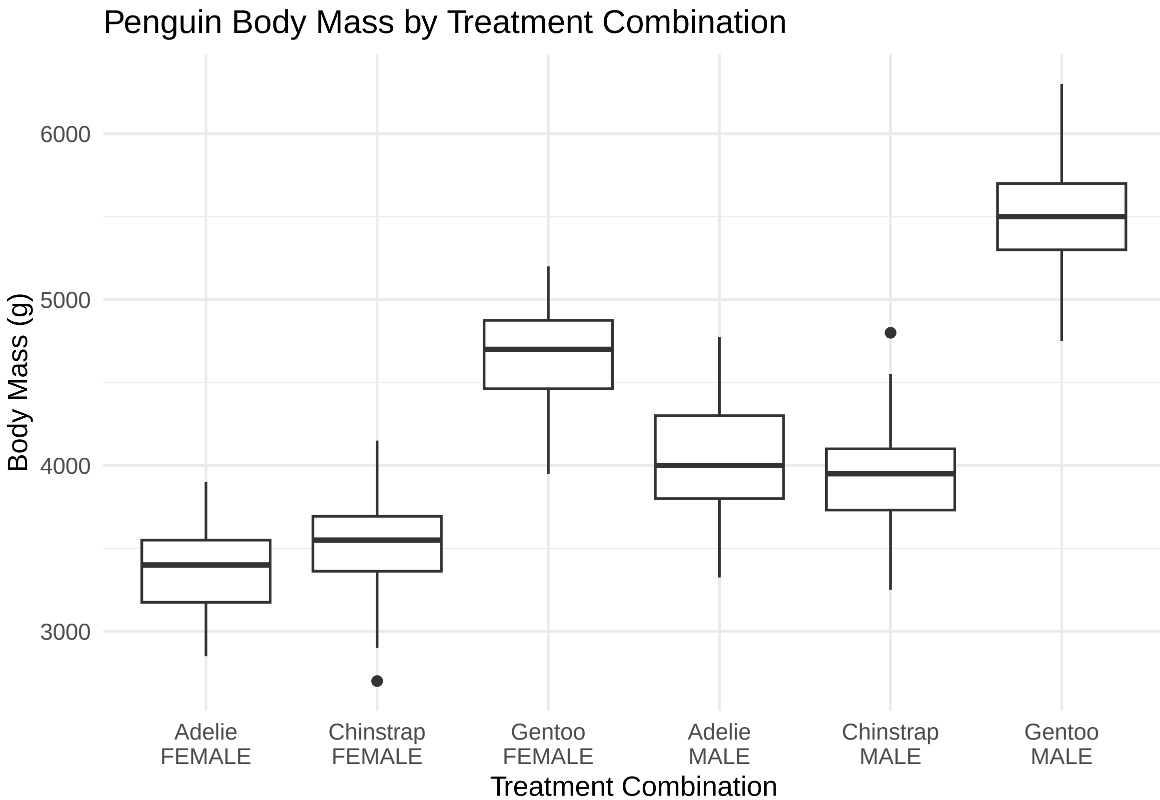
3.1.3.2 Visualization

3.1.3.2.1 Histogram of Penguin Body Mass



The histogram shows that penguin body masses range from approximately 2700 g to 6300 g, with most observations concentrated between 3000 g and 5000 g. The distribution shows a mild right skew, with a longer tail toward higher body masses. This is consistent with the mean body mass (4207.06 g) being higher than the median (4050 g). The distribution also shows relatively high frequencies around 3500 to 4000 g, while fewer observations are found at the upper end of the body mass range.

3.1.3.2.2 Box Plot

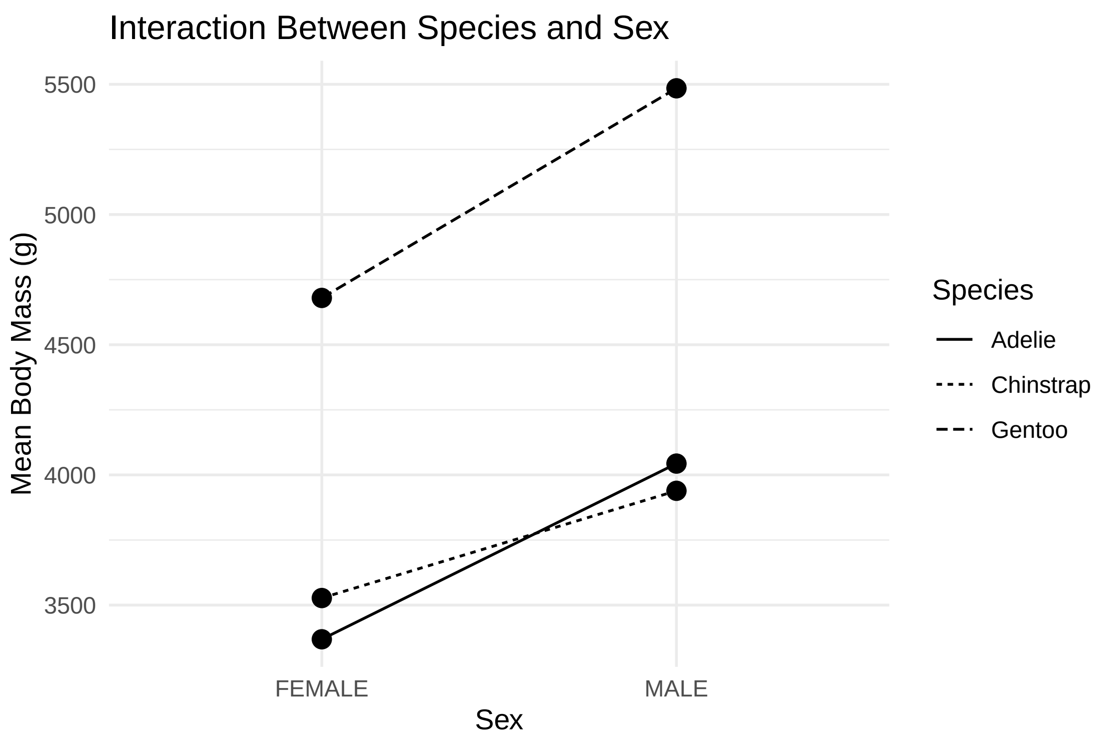


The box plot compares the distribution of penguin body mass across the six treatment combinations of species and sex. Male penguins generally have higher median and mean body mass than female penguins within each species.

The difference between male and female body mass appears across all three species. The largest difference is observed for Adelie penguins, with mean body mass increasing from 3368.84 g for females to 4043.49 g for males. For Chinstrap penguins, the corresponding means are 3527.21 g and 3938.97 g, while for Gentoo penguins they are 4679.74 g and 5484.84 g.

The spread of the groups is not completely uniform, with some groups showing wider variability than others. A few potential low outliers are also visible, particularly in the Chinstrap female group. Overall, the plot suggests that both species and sex may be associated with penguin body mass, while the differences in the male-female effect across species provide descriptive evidence of a possible interaction between the two factors.

3.1.3.2.3 Interaction Plot



The interaction plot shows the mean body mass for each combination of species and sex. Across all three species, male penguins have higher mean body mass than female penguins. The mean body mass increases by approximately 674.66 g for Adelie, 411.76 g for Chinstrap, and 805.10 g for Gentoo penguins.

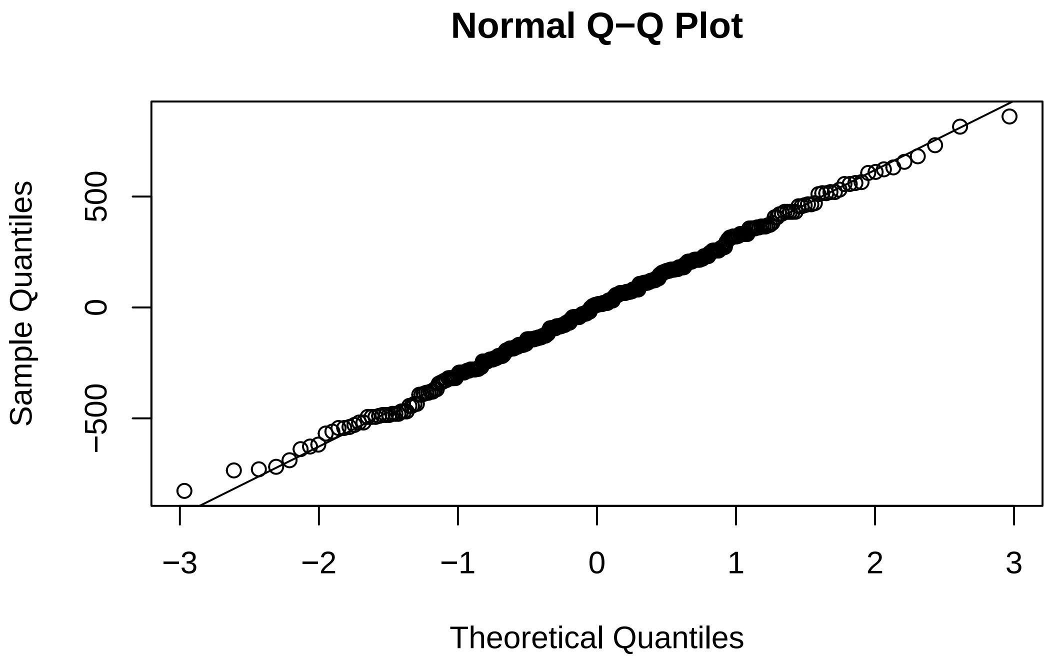
The lines are not perfectly parallel. In particular, the Adelie and Chinstrap lines cross between the female and male groups, while the Adelie and Gentoo lines show relatively similar slopes. This provides some descriptive evidence that the difference in body mass between males and females may vary across species.

However, the interaction pattern appears relatively modest overall, and the plot alone cannot determine whether the interaction is statistically significant. The significance of the species-by-sex interaction will be formally assessed using the two-way ANOVA.

3.1.3.3 Assumptions Check

```
qqnorm(residual)
qqline(residual)
```

3.1.3.3.1 Normality of Residuals




```r
shapiro.test(residual)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residual
## W = 0.99776, p-value = 0.9367
```

The normality assumption was assessed using the normal Q-Q plot and the Shapiro-Wilk test. The hypotheses for the Shapiro-Wilk test are:

$$\begin{cases} H_0 : \text{The residuals follow a normal distribution} \\ H_1 : \text{The residuals do not follow a normal distribution} \end{cases}$$

The Shapiro-Wilk test resulted in

$$W = 0.99776, \qquad p = 0.9367.$$

Since the p-value is greater than 0.05, we fail to reject H_0. Therefore, there is no statistical evidence that the ANOVA residuals deviate from a normal distribution.

The Q-Q plot also shows that the residuals generally follow the reference line, with only minor deviations at the tails. The Shapiro-Wilk test gives $W = 0.99776$ with a p-value of 0.9367, providing further support for the normality assumption.

Overall, the residuals show no substantial departure from normality, and the normality assumption is considered satisfied for the subsequent two-way ANOVA.

```r
leveneTest(
  body_mass_g ~
    species * sex,
  data = analysis_data
)
```

3.1.3.3.2 Homogeneity of Variances

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   5  1.3908 0.2272
##       327
```

The homogeneity of variance assumption was assessed using the treatment group box plots and Levene's test. The six treatment combinations are formed by `species` and `sex`.

The hypotheses are:

$$\begin{cases} H_0 : \sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \cdots = \sigma_6^2 \\ H_1 : \text{At least one treatment group has a different variance} \end{cases}$$

Levene's test produced

$$F(5, 327) = 1.3908, \qquad p = 0.2272$$

Since the p-value is greater than 0.05, we fail to reject H_0. Therefore, there is no statistical evidence that the variances differ across the six treatment combinations.

The group variances range from approximately 72,565.64 to 120,278.25, with the Adelie male group having the largest variance and the Adelie female group having the smallest variance. Although the group variances are not identical, Levene's test suggests that the differences are not statistically significant. Therefore, the homogeneity of variance assumption is considered satisfied for the subsequent two-way ANOVA.

```r
sum(duplicated(analysis_data))
```

3.1.3.3.3 Independence of Errors

```
## [1] 169
```

The independence assumption requires that the error terms are independent:

$$\epsilon_{ijk} \overset{ind}{\sim} N(0, \sigma^2).$$

The independence assumption was assessed based on the structure of the dataset. Each observation represents an individual penguin, and the dataset does not contain repeated measurements or a time-dependent structure. Although some observations have identical values for the selected variables (`body_mass_g`, `species`, and `sex`), these do not necessarily represent duplicate penguins, since different penguins can naturally have the same body mass and characteristics.

Therefore, independence is considered reasonable based on the available dataset information. However, it cannot be formally verified because the dataset does not provide a unique identifier or detailed information about the sampling procedure.

3.1.3.4 Overall Assessment

The independence assumption is considered reasonable based on the structure of the dataset. Each observation represents an individual penguin, and there is no time-dependent or repeated-measurement structure identified in the analysis data. However, independence cannot be formally verified because the dataset does not provide a unique identifier or detailed information about the sampling procedure.

The normality assumption is considered satisfied based on both the Q-Q plot and the Shapiro-Wilk test ($p = 0.9367$), which provides no statistical evidence of a departure from normality. The homogeneity of variance assumption is also considered satisfied based on the non-significant Levene's test ($p = 0.2272$).

Overall, the assumptions are considered reasonably adequate for conducting the two-way ANOVA. No substantial concerns are observed regarding normality or homogeneity of variance, while the independence assumption is considered reasonable based on the available dataset information.

3.2 ANOVA / Factorial design analysis

```
model <- aov(body_mass_g ~ species * sex ,data=analysis_data)

anova_table <- as.data.frame(summary(model)[[1]])
p_value <- anova_table$`Pr(>F)`
f_value <- anova_table$`F value`

kable(anova_table, digits=3, caption = "ANOVA result")
```

Table 15: ANOVA result

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
species	2	145190219	72595109.56	758.358	0
sex	1	37090262	37090261.78	387.460	0
species:sex	2	1676557	838278.37	8.757	0
Residuals	327	31302628	95726.69	NA	NA

Using significant level α of 0.05, we observe that all these p-value are less than α.

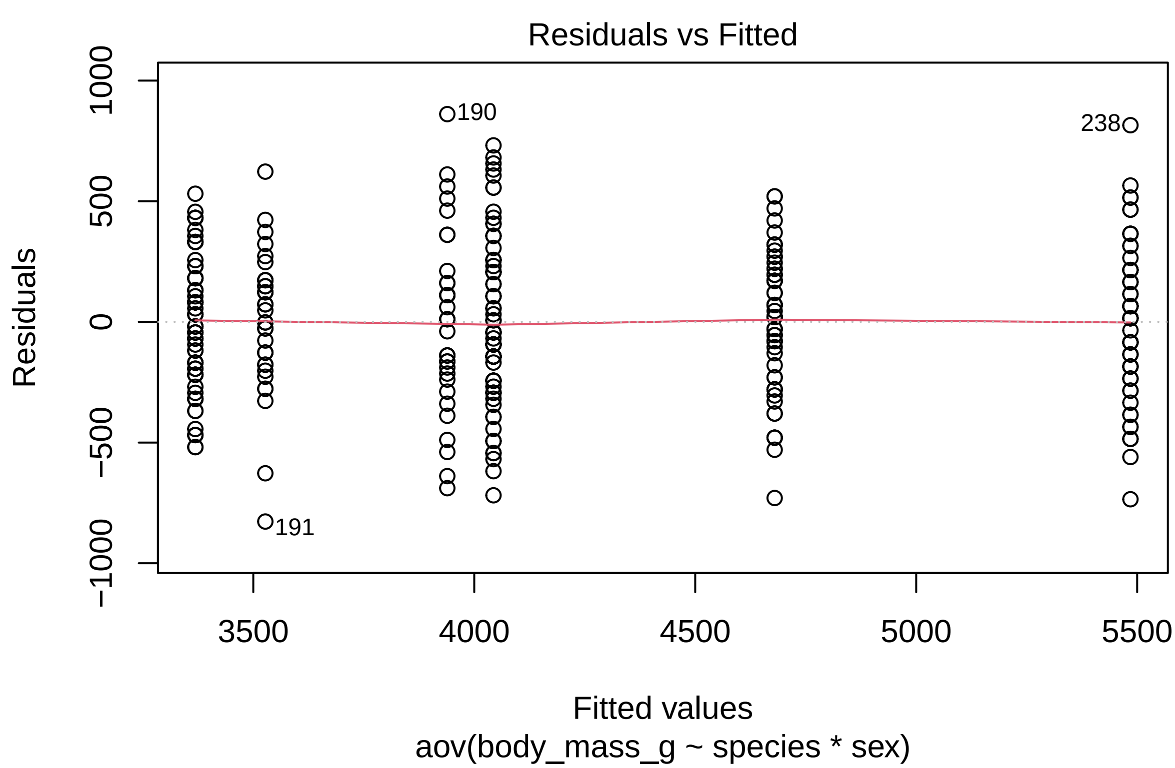
For species, the sum of squares is 145,190,219, which is the largest among all factors. Together with the very large F-value of 758.358 and $p < 0.001$, this indicates that species explains a very large amount of the variation in the response variable and has a highly significant effect.

For sex, the sum of squares is 37,090,262, which is also large compared with the residual sum of squares. The F-value of 387.460 and $p < 0.001$ show that sex has a highly significant effect on the response variable.

For the species - sex interaction, the degrees of freedom is 2, because the interaction involves the three species and two sex groups. Its sum of squares is 1,676,557, which is much smaller than the sums of squares for species and sex. However, the interaction is still statistically significant, with an F-value of 8.757 and $p < 0.001$. This indicates that although the interaction explains less variation, the effect of sex still differs significantly among species.

Overall, species contributes the largest amount of variation, as shown by its highest Sum Sq (145,190,219) and F-value (758.358). Sex also contributes a substantial amount of variation, with a Sum Sq of 37,090,262 and F-value of 387.460. The interaction has a much smaller Sum Sq (1,676,557) and F-value (8.757), but it remains significant. Therefore, all three effects—species, sex, and species x sex are statistically significant ($p < 0.001$), with species showing the strongest effect.

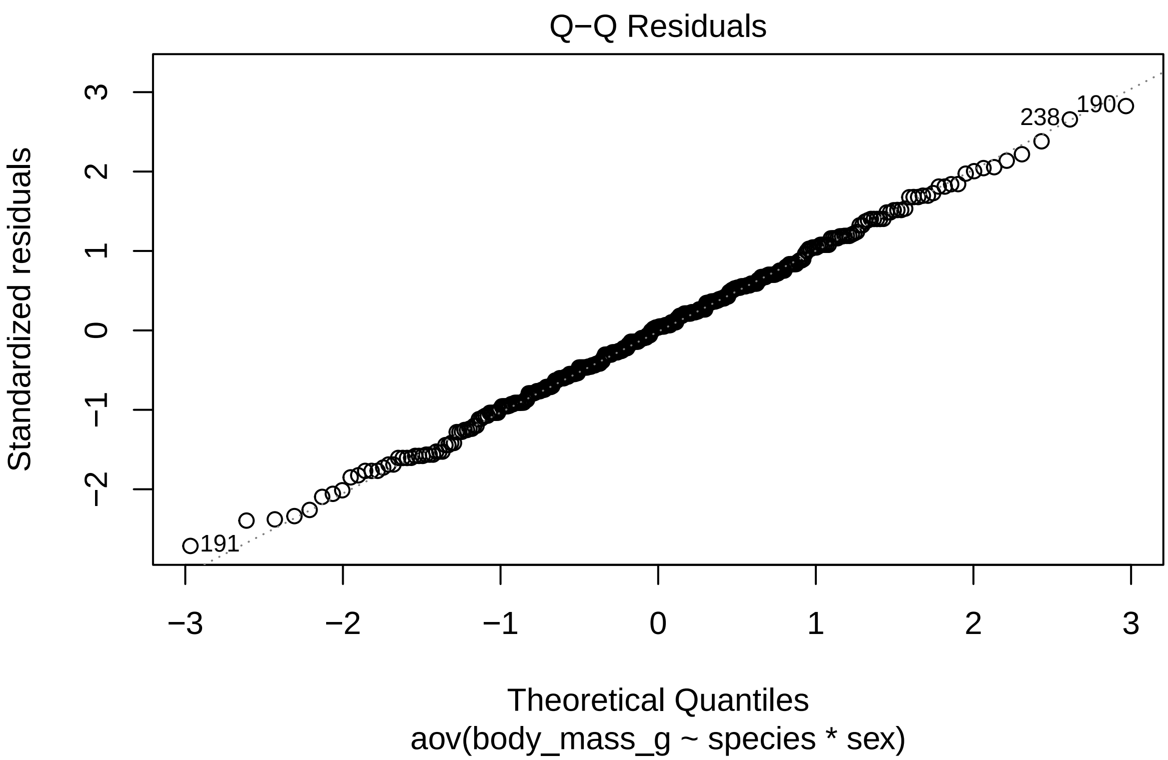
```r
plot(model, which = 1)
```



This residuals are generally scattered around the horizontal zero line across the different fitted values, and the red trend line remains close to zero. This indicates that there is no strong systematic pattern or curvature in the residuals, suggesting that the ANOVA model captures the main structure of the data reasonably well.

The spread of the residuals is somewhat different between the fitted-value groups, but there is no clear funnel-shaped pattern where the spread consistently increases or decreases as the fitted values increase. Therefore, there is no strong visual evidence of severe heteroscedasticity from this plot.

```r
plot(model, which = 2)
```



This Q–Q plot shows most of the points closely follow the diagonal reference line, particularly in the middle section of the plot, indicating that the residuals are approximately normally distributed. This suggests that the assumption of normality required for ANOVA is reasonably satisfied.

3.3 Posthoc analysis

```r
tukey <- TukeyHSD(model, which = "species")
kable(tukey$species, digits = 3, caption = "Tukey HSD Results")
```

Table 16: Tukey HSD Results

	diff	lwr	upr	p adj
Chinstrap-Adelie	26.924	-80.026	133.874	0.824
Gentoo-Adelie	1386.273	1296.307	1476.238	0.000
Gentoo-Chinstrap	1359.349	1248.611	1470.087	0.000

The comparison between Chinstrap and Adelie is not statistically significant, with an adjusted p-value of 0.824, which is greater than 0.05. The confidence interval also includes zero (-80.026 to 133.874). Therefore, there is insufficient evidence to conclude that the mean response differs between Chinstrap and Adelie.

In contrast, Gentoo differs significantly from Adelie, with a mean difference of 1386.273 and an adjusted p-value less than 0.001. The 95% confidence interval (1296.307 to 1476.238) does not contain zero, providing strong evidence that the two species have different mean responses. Since the mean difference is positive, the mean response for Gentoo is higher than Adelie by approximately 1386.273 units.

Similarly, Gentoo differs significantly from Chinstrap, with a mean difference of 1359.349 and an adjusted p-value less than 0.001. The 95% confidence interval (1248.611 to 1470.087) is entirely positive, indicating that the mean response for Gentoo is higher than Chinstrap by approximately 1359.349 units.

Overall, the Tukey HSD results show that Gentoo is significantly different from both Adelie and Chinstrap, whereas Adelie and Chinstrap do not differ significantly from each other. This provides more specific information about the significant species effect identified by the ANOVA.

```r
sex_analysis <- t.test(body_mass_g ~ sex,data=analysis_data)

tidy_t_test <- tidy(sex_analysis)

kable(
  tidy_t_test[, c("estimate", "statistic", "p.value", "conf.low", "conf.high")],
  digits = 3,
  col.names = c("Mean Difference", "t-Statistic", "p-value", "Lower CI", "Upper
   ↪  CI"),
  caption = "Independent t-test Comparing Penguin Body Mass by Sex"
)
```

Table 17: Independent t-test Comparing Penguin Body Mass by Sex

Mean Difference	t-Statistic	p-value	Lower CI	Upper CI
-683.412	-8.555	0	-840.578	-526.245

An independent-samples t-test was conducted to compare penguin body mass between male and female penguins. The results showed a mean difference of -683.412 grams, with a t-statistic of -8.555 and a p-value of < 0.001.

The very small p-value indicates a statistically significant difference in body mass between the two sexes. The mean difference of -683.412 grams indicates that the first sex group in the comparison has an average body mass approximately 683.4 grams lower than the second sex group.

The 95% confidence interval for the mean difference ranges from -840.578 to -526.245 grams. Since the entire interval is below zero, the estimated difference is consistently in the same direction. Therefore, the results indicate that one sex has substantially greater average body mass than the other.

Overall, sex is associated with a significant difference in penguin body mass, with an average difference of approximately 683 grams between the two sexes.

```
emm_result <- emmeans(model, pairwise ~ sex | species, adjust = "tukey")

kable(
  as.data.frame(emm_result$contrasts),
  digits = 3,
  col.names = c("Contrast", "Species", "Difference (g)", "Std. Error", "df",
  ↪  "t-ratio", "p.value"),
  align = "llrrrrr",
  caption = "Pairwise Comparisons of Body Mass Between Sexes Stratified by
  ↪  Penguin Species"
)
```

Table 18: Pairwise Comparisons of Body Mass Between Sexes Stratified by Penguin Species

Contrast	Species	Difference (g)	Std. Error	df	t-ratio	p.value
FEMALE - MALE	Adelie	-674.658	51.212	327	-13.174	0
FEMALE - MALE	Chinstrap	-411.765	75.040	327	-5.487	0
FEMALE - MALE	Gentoo	-805.095	56.743	327	-14.188	0

Pairwise comparisons showed that male penguins had significantly higher body mass than females in all three species ($p<0.001$). The differences were approximately 674.7 g for Adelie, 411.8 g for Chinstrap, and 805.1 g for Gentoo. The largest sex difference occurred in Gentoo, while the smallest occurred in Chinstrap.

3.3.1 Practical significance

The results are not only statistically significant but also practically meaningful. The differences in body mass between species are large, particularly between Gentoo and the other two species, with differences of more than 1,300 g. The sex differences are also substantial, ranging from approximately 412 g to 805 g, with males consistently heavier than females.

Therefore, species and sex should be considered when assessing or comparing penguin body mass. Ignoring these factors could lead to misleading comparisons, especially when comparing groups with different species or sex compositions.

3.4 Conclusion and evaluation

The two-way ANOVA showed that species and sex both have significant effects on penguin body mass ($p<0.001$). The Species x Sex interaction was also significant ($p<0.001$), indicating that the effect of sex on body mass varies among species. Tukey's HSD showed that Gentoo differs significantly from both Adelie and Chinstrap, while Adelie and Chinstrap do not differ significantly. Pairwise comparisons further showed that males have significantly higher body mass than females in all three species, with the largest difference observed in Gentoo.

The analysis provides strong evidence of differences in body mass by species and sex. However, the ANOVA assumptions should be considered, particularly normality and homogeneity of variance. The data may also be unbalanced across groups, which can affect the analysis. Future analysis could use robust methods or transformations if assumption violations are substantial.

Based on these results, species and sex should both be considered when analyzing or predicting penguin body mass. In particular, the significant interaction suggests that the effect of sex should not be interpreted independently of species. Further studies could collect more balanced samples across species and sexes to improve the reliability and generalizability of the findings.